

Sequential Extremal Principle: Refinements and Applications

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Dedicated to the memory of Prof. Franco Giannessi and his great contributions to optimisation and the journal

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Abstract Sequential extremality and stationarity properties are discussed with the emphasis on those corresponding to fixed sequences of translations. Exact quantitative characterisations of the properties are provided. We show, in particular, that the sequential (as well as conventional) extremality and approximate stationarity properties possess certain stability, while the (non-approximate) stationarity does not. Dual necessary conditions for the sequential extremality and stationarity properties with fixed and non-fixed sequences of translations are established. A version of the sequential extended extremal principle is formulated. In the statements, we employ certain generalised separation conditions (**GS**) and (**GS** _{α}) as well as a complementary primal-dual condition (**PD**). To illustrate the model, we prove dual optimality/stationarity conditions for a constrained minimisation problem in which the minimal value is not necessarily attained.

Keywords extremal principle · separation · stationarity · transversality · optimality conditions

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1 Introduction

We establish some refinements of the *sequential extremal principle* introduced recently in [10] as an extension of the conventional extremal principle originally proposed in [24] and further developed in [27,30] and many other publications; see, e.g., [4,9,13,19,26,31,36]. Being natural generalisations of the classical *separation theorem*, extremal principles (generalised separation results) are widely used in variational analysis and naturally translate into

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necessary optimality conditions (multiplier rules) and various subdifferential, normal cone and coderivative calculus results in nonconvex settings; in addition to the above references, we refer the reader to [1, 15, 16, 18, 28, 29].

The conventional extremal principle gives dual necessary conditions, which can be interpreted as generalised separation, for the extremal behaviour of a finite collection of sets near a given point in their intersection (often referred to as *extremal point*). It was shown in [16, 17] (see also [3, 18–21, 23]) that the generalised separation conditions can be true under weaker than extremality assumptions which can be interpreted as some kinds of *stationarity*.

The next definition introduces the mentioned properties. Here and throughout the paper, we consider a collection of $n > 1$ nonempty subsets $\Omega_1, \dots, \Omega_n$ of a normed vector space X and write $\{\Omega_1, \dots, \Omega_n\}$ to denote the collection of these sets as a single object. Symbols $\mathbb{B}_X$ and $B_\rho(\bar{x})$ denote the open unit ball and open ball with centre $\bar{x}$ and radius $\rho > 0$, respectively, while $\mathbb{B}_X$ denotes the closed unit ball.

Definition 1.1 (Extremality, stationarity and approximate stationarity) Let $\bar{x} \in \bigcap_{i=1}^n \Omega_i$. The collection $\{\Omega_1, \dots, \Omega_n\}$ is

- (i) extremal at $\bar{x}$ if there is a $\rho > 0$ such that, for any $\varepsilon > 0$, there exist $a_1, \dots, a_n \in \varepsilon \mathbb{B}_X$ such that $\bigcap_{i=1}^n (\Omega_i - a_i) \cap B_\rho(\bar{x}) = \emptyset$;
- (ii) stationary at $\bar{x}$ if, for any $\varepsilon > 0$, there exist a $\rho \in (0, \varepsilon)$ and $a_1, \dots, a_n \in \varepsilon \rho \mathbb{B}_X$ such that $\bigcap_{i=1}^n (\Omega_i - a_i) \cap B_\rho(\bar{x}) = \emptyset$;
- (iii) approximately stationary at $\bar{x}$ if, for any $\varepsilon > 0$, there exist a $\rho \in (0, \varepsilon)$, $x_i \in \Omega_i \cap B_\varepsilon(\bar{x})$ and $a_i \in \varepsilon \rho \mathbb{B}_X$ ($i = 1, \dots, n$) such that $\bigcap_{i=1}^n (\Omega_i - x_i - a_i) \cap (\rho \mathbb{B}_X) = \emptyset$.

The relationships between the properties in Definition 1.1 are straightforward: (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii). In Asplund spaces, the approximate stationarity, the weakest of the three properties, is equivalent to the dual necessary conditions in the conventional extremal principle, i.e., the *extended extremal principle* holds; cf. [18, Theorem 3.7].

Theorem 1.1 (Extended extremal principle) Let X be Asplund, $\Omega_1, \dots, \Omega_n$ be closed, and $\bar{x} \in \bigcap_{i=1}^n \Omega_i$. The collection $\{\Omega_1, \dots, \Omega_n\}$ is approximately stationary at $\bar{x}$ if and only if, for any $\varepsilon > 0$, there exist $x_i \in \Omega_i \cap B_\varepsilon(\bar{x})$ and $x_i^* \in N_{\Omega_i}^F(x_i)$ ($i = 1, \dots, n$) such that $\|\sum_{i=1}^n x_i^*\| < \varepsilon$ and $\sum_{i=1}^n \|x_i^*\| = 1$.

Recall that a Banach space is Asplund if every continuous convex function on an open convex set is Fréchet differentiable on a dense subset, or equivalently, if the dual of each its separable subspace is separable. We refer the reader to [1, 27, 34] for discussions about and characterisations of Asplund spaces. Theorem 1.1 employs *Fréchet normal cones*. The dual conditions are formulated in a *fuzzy* form and can be interpreted as *generalised separation*; cf. [4]. The conventional extremal principle [24, 27, 30], claiming that these generalised separation conditions are implied by $\{\Omega_1, \dots, \Omega_n\}$ being extremal at $\bar{x}$, is an immediate consequence of Theorem 1.1.

The extremality and stationarity concepts in Definition 1.1 and their characterisations in the conventional and extended extremal principle as well as other typical generalisations and extensions are attached to a fixed point in the intersection of the sets (modeling an optimal or stationary point). As observed in [10], despite its recognized versatility and numerous applications, this model does not cover an important class of optimisation problems and calculus relations ‘at infinity’ [14, 32, 33] involving unbounded sets, where we can only have minimising (or stationary in some sense) sequences; see [10, Examples 1.5, 2.2, 4.4 and 4.7–4.9].

It was suggested in [10] to consider sequential versions of the properties in Definition 1.1 replacing the point $\bar{x} \in \bigcap_{i=1}^n \Omega_i$ (which may not exist) by appropriate sequences. Below is a slightly simplified version of [10, Definition 2.1].

Definition 1.2 (Sequential extremality, stationarity and approximate stationarity) The collection $\{\Omega_1, \dots, \Omega_n\}$ is

(i) extremal at sequences $\{x_i^k\} \subset \Omega_i$ ($i = 1, \dots, n$) if

$$\text{diam}\{x_1^k, \dots, x_n^k\} := \max_{1 \leq i, j \leq n} \|x_i - x_j\| \rightarrow 0 \quad \text{as } k \rightarrow +\infty, \quad (1)$$

and there is a $\rho > 0$ such that, for any $\varepsilon > 0$, there exist an integer $k > \varepsilon^{-1}$ and points $a_1, \dots, a_n \in \varepsilon \mathbb{B}_X$ such that $\bigcap_{i=1}^n (\Omega_i - x_i^k - a_i) \cap (\rho \mathbb{B}_X) = \emptyset$;

(ii) stationary at sequences $\{x_i^k\} \subset \Omega_i$ ($i = 1, \dots, n$) if condition (1) is satisfied, and, for any $\varepsilon > 0$, there exist an integer $k > \varepsilon^{-1}$, a $\rho \in (0, \varepsilon)$, and points $a_1, \dots, a_n \in \varepsilon \rho \mathbb{B}_X$ such that $\bigcap_{i=1}^n (\Omega_i - x_i^k - a_i) \cap (\rho \mathbb{B}_X) = \emptyset$;

(iii) approximately stationary at a sequence $\{x^k\} \subset X$ if, for any $\varepsilon > 0$, there exist an integer $k > \varepsilon^{-1}$, a $\rho \in (0, \varepsilon)$, and points $x_i \in \Omega_i \cap B_\varepsilon(x^k)$ and $a_i \in \varepsilon \rho \mathbb{B}_X$ ($i = 1, \dots, n$) such that $\bigcap_{i=1}^n (\Omega_i - x_i - a_i) \cap (\rho \mathbb{B}_X) = \emptyset$.

Definition 1.1 is a particular case of Definition 1.2 with $x_i^k := \bar{x}$ for all $i = 1, \dots, n$ and $k \in \mathbb{N}$ in parts (i) and (ii), and $x^k := \bar{x}$ for all $k \in \mathbb{N}$ in part (iii). Here, the sets $\Omega_1, \dots, \Omega_n$ are not supposed to have a common point. Instead, condition (1) in parts (i) and (ii), and conditions $x_i \in \Omega_i \cap B_\varepsilon(x^k)$ ($i = 1, \dots, n$) in part (iii) of Definition 1.2 ensure that the (generalised) distance between the sets is 0. The sequences in the definition may be unbounded. Similarly to Definition 1.1, it holds (i) $\Rightarrow$ (ii) in Definition 1.2 and, if $\{\Omega_1, \dots, \Omega_n\}$ is stationary at sequences $\{x_i^k\} \subset \Omega_i$ ($i = 1, \dots, n$), then, for each $i \in \{1, \dots, n\}$, it is approximately stationary at the sequence $\{x_i^k\}$.

Employing the generalised separation result from [9, Theorem 3.1] (proved using standard techniques based on the Ekeland variational principle and subdifferential sum rules), dual necessary conditions for the sequential approximate stationarity (and for the more advanced sequential approximate α -stationarity) were established in [10]. These conditions preserve the generalised separation pattern typical of the conventional extremal principle. In the Asplund space setting, they lead to the following statement generalising and extending Theorem 1.1; cf. [10, Corollary 3.7].

Theorem 1.2 (Sequential extended extremal principle) Let X be Asplund, $\Omega_1, \dots, \Omega_n$ be closed, and $\{x^k\} \subset X$. The collection $\{\Omega_1, \dots, \Omega_n\}$ is approximately stationary at $\{x^k\}$ if and only if, for any $\varepsilon > 0$, there exist an integer $k > \varepsilon^{-1}$, and points $x_i \in \Omega_i \cap B_\varepsilon(x^k)$ and $x_i^* \in N_\Omega^F(x_i)$ ($i = 1, \dots, n$) such that $\|\sum_{i=1}^n x_i^*\| < \varepsilon$ and $\sum_{i=1}^n \|x_i^*\| = 1$.

Remark 1.1 Compared to Theorem 1.2, the statement of [10, Corollary 3.7] contains an additional dual characterisation of sequential approximate stationarity of $\{\Omega_1, \dots, \Omega_n\}$ at $\{x^k\}$: for any $\varepsilon > 0$ and $\tau \in (0, 1)$, there exist an integer $k > \varepsilon^{-1}$, and points $x_0 \in \varepsilon \mathbb{B}_X$, $x_i, x'_i \in \Omega_i \cap B_\varepsilon(x^k)$, $a_i \in \varepsilon \mathbb{B}_X$, $x_i^* \in N_\Omega^F(x_i)$ ($i = 1, \dots, n$) such that $\|\sum_{i=1}^n x_i^*\| < \varepsilon$, $\sum_{i=1}^n \|x_i^*\| = 1$, and

$$\sum_{i=1}^n \langle x_i^*, x_0 + a_i + x'_i - x_i \rangle > \tau \max_{1 \leq i \leq n} \|x_0 + a_i + x'_i - x_i\|.$$

The origins of this type of conditions can be traced back to the *unified separation theorem* by Zheng & Ng [36].

The above characterisation obviously implies the one in Theorem 1.2. At the same time, it is a straightforward consequence of the one in Theorem 1.2. Indeed, let $\varepsilon > 0$, $k \in \mathbb{N}$, $x_i \in \Omega_i \cap B_\varepsilon(x^k)$, $x_i^* \in N_\Omega^F(x_i)$ ($i = 1, \dots, n$), $\|\sum_{i=1}^n x_i^*\| < \varepsilon$ and $\sum_{i=1}^n \|x_i^*\| = 1$. Let $\tau \in (0, 1)$ and $\tau' \in (\tau, 1)$. For each $i = 1, \dots, n$, there exists an $a'_i \in X$ such that $\|a'_i\| = 1$ and $\langle x_i^*, a'_i \rangle \geq \tau' \|x_i^*\|$. Choose an $\varepsilon' \in (0, \varepsilon)$ and define $a_i := \varepsilon' a'_i$. Then $\|a_i\| = \varepsilon' < \varepsilon$ ($i = 1, \dots, n$), and

$$\sum_{i=1}^n \langle x_i^*, a_i \rangle \geq \varepsilon' \tau' \sum_{i=1}^n \|x_i^*\| = \varepsilon' \tau' > \tau \max_{1 \leq i \leq n} \|a_i\|.$$

Thus, the characterisation formulated above holds with $x_0 := 0$ and $x'_i := x_i$ ($i = 1, \dots, n$).

The sequential model developed in [10] extends the applicability of the approach based on extremal principles. In particular, it leads to natural dual optimality conditions (multiplier rules) for minimising sequences; see [10, Section 5]. This sequential model is more flexible than the model proposed in [14,32,33] and does not require introducing an artificial infinity point.

In this paper, we extend and improve Definition 1.2 along several lines.

- (i) The model adopted in the current paper does not assume condition (1) making an essential part of the properties in parts (i) and (ii) of the definition, and implicitly present also in part (iii). This natural condition is satisfied in many settings involving unbounded sequences modeling optimisation problems ‘at infinity’ as discussed in [10]. At the same time, it is superfluous when proving dual necessary conditions. In the conventional ‘at-a-point’ setting, this fact was observed in [3, Section 4.2].
- (ii) The properties in Definition 1.2 are defined for the given sequences $\{x_i^k\} \subset \Omega_i$ ($i = 1, \dots, n$) in parts (i) and (ii), and $\{x^k\} \subset X$ in part (iii). However, a closer look at the definition shows that the properties are actually determined by some subsequences of the given sequences. The definitions adopted in the current paper are more specific in identifying the sequences relevant for the properties.
- (iii) The last property in Definition 1.2 is defined using a slightly different language compared to the first two properties. This complicates the comparison of the definitions and results. One can see this already when reading the above items. An advantage of the model adopted in the current paper is the uniformity of the presentation of all three groups of the studied properties: extremality, stationarity and approximate stationarity, simplifying the comparison of the definitions and results.
- (iv) All three properties in Definition 1.2 (as well as Definition 1.1) implicitly assume the existence of certain sequences of translations converging to zero. These sequences play an important role in the definitions. In the current paper, we make them explicit and study the properties corresponding to fixed sequences of translations. One can go further and consider more general than translations deformations of the given sets; cf. [11,12].
- (v) Unlike the descriptive Definition 1.2, the corresponding properties are defined in this paper using exact quantitative estimates of the properties; see Definition 2.1.

These modifications lead to a clearer model resulting in improved sequential dual necessary conditions with some additional restrictions on the dual variables and also shed some new light on extremality/stationarity settings, especially on the role of the vectors/sequences of translations.

The new definitions are introduced and discussed in Section 2. It consists of two subsections. In the first one, we discuss the properties corresponding to fixed sequences of translations $\{a_i^k\} \subset X$ ($i = 1, \dots, n$). This line of research is new even in the conventional ‘at-a-point’ setting. The exact quantitative characterisations of the respective properties are provided. We show, in particular, that the sequential (as well as conventional) extremality and approximate stationarity properties possess certain stability, while the (non-approximate) stationarity does not. The established relations are then used in the second subsection when proving certain facts for the properties defined for *some* sequences $\{a_i^k\}$ ($i = 1, \dots, n$). The conventional setting when the sequences $\{x_i^k\} \subset \Omega_i$ ($i = 1, \dots, n$) reduce to a single point in the intersection of the sets is given a special attention.

In Section 3, we establish dual necessary conditions for the sequential extremality and stationarity properties discussed in Section 2 and a more elaborate version of the sequential extended extremal principle. In the statements, we employ certain generalised separation conditions (GS) and (GS $_{\alpha}$) as well as a complementary primal-dual condition (PD) providing

additional restrictions on the associated dual vectors. When proving the dual necessary conditions, we employ a simplified version of [9, Corollary 3.2] instead of the general separation result [9, Theorem 3.1] used in [10], producing a straightforward proof.

To illustrate the model, we consider in Section 4 a constrained minimisation problem in which the minimal value is not necessarily attained and deduce stronger than in [10] sequential optimality and stationarity conditions for more general types of stationary sequences. Some concluding remarks are collected in Section 5.

Preliminaries

Our basic notation is standard; see, e.g., [1, 27]. Throughout the paper, X is a normed space (typically Banach or Asplund). Its topological dual is denoted by X^* , while $\langle \cdot, \cdot \rangle$ denotes the bilinear form defining the pairing between the two spaces. The product spaces X^n and $X \times \mathbb{R}$ are assumed to be equipped with the maximum norms. We use the same symbols $\|\cdot\|$ and $d(\cdot, \cdot)$ to denote norms and distances (including point-to-set distances with the convention $d(x, \emptyset) = +\infty$ for any x) in all (primal and dual) spaces. The results of the current paper can be easily extended to the setting of general product norms satisfying certain compatibility conditions with the norm on X (particularly, ℓ^p norms). This can only affect some quantitative estimates. We refer the readers to [7, 9, 10] for the details. Note that the counterparts of Definition 1.2 and Theorem 1.2 are formulated in [10] using general product norms.

Symbols $\mathbb{R}$, $\mathbb{R}_+$ and $\mathbb{N}$ stand for the sets of all, respectively, real, nonnegative real and positive integer numbers. If $\alpha \in \mathbb{R}$, then $\alpha_+ := \max\{\alpha, 0\}$ is the *positive part* of α . The notation $\{x^k\} \subset \Omega$ denotes a sequence of points $x^k \in \Omega$ ($k \in \mathbb{N}$). We sometimes use a shorter notation $\sigma_x := \{x^k\}$.

Normal cones and subdifferentials. We first recall the definitions of normal cones and subdifferentials in the sense of Fréchet and Clarke; see, e.g., [5, 18, 27]. Given a subset Ω of a normed space X and a point $\bar{x} \in \Omega$, the sets

$$N_{\Omega}^F(\bar{x}) := \left\{ x^* \in X^* \mid \limsup_{\Omega \ni x \rightarrow \bar{x}, x \neq \bar{x}} \frac{\langle x^*, x - \bar{x} \rangle}{\|x - \bar{x}\|} \leq 0 \right\}, \quad (2)$$

$$N_{\Omega}^C(\bar{x}) := \left\{ x^* \in X^* \mid \langle x^*, z \rangle \leq 0 \text{ for all } z \in T_{\Omega}^C(\bar{x}) \right\} \quad (3)$$

are the, respectively, *Fréchet* and *Clarke normal cones* to Ω at $\bar{x}$. Symbol $T_{\Omega}^C(\bar{x})$ in (3) stands for the *Clarke tangent cone* to Ω at $\bar{x}$. The sets (2) and (3) are nonempty closed convex cones satisfying $N_{\Omega}^F(\bar{x}) \subset N_{\Omega}^C(\bar{x})$. If Ω is a convex set, they reduce to the normal cone $N_{\Omega}(\bar{x})$ in the sense of convex analysis.

For an extended-real-valued function $f : X \rightarrow \mathbb{R}_{\infty} := \mathbb{R} \cup \{+\infty\}$ on a normed space X with epigraph $\text{epi } f := \{(x, \alpha) \in X \times \mathbb{R} \mid f(x) \leq \alpha\}$, the *Fréchet* and *Clarke subdifferentials* of f at $\bar{x} \in \text{dom } f := \{x \in X \mid f(x) < +\infty\}$ are defined, respectively, by

$$\partial^F f(\bar{x}) := \left\{ x^* \in X^* \mid (x^*, -1) \in N_{\text{epi } f}^F(\bar{x}, f(\bar{x})) \right\}, \quad (4)$$

$$\partial^C f(\bar{x}) := \left\{ x^* \in X^* \mid (x^*, -1) \in N_{\text{epi } f}^C(\bar{x}, f(\bar{x})) \right\}. \quad (5)$$

The sets (4) and (5) are closed and convex, and satisfy $\partial^F f(\bar{x}) \subset \partial^C f(\bar{x})$. If f is convex, they reduce to the subdifferential $\partial f(\bar{x})$ in the sense of convex analysis.

We often use the generic notations N and ∂ with the convention that $N := N^C$ and $\partial := \partial^C$ if X is a general normed space, and $N := N^F$ and $\partial := \partial^F$ if X is Asplund.

Generalised separation. The proof of our main result is based on the next generalised separation statement. It is a simplified version of a more general result from [9, Corollary 3.2] extending the *unified separation theorems* by Zheng & Ng [35,36] and their slightly more advanced versions in [11, Lemma 2.1].

Theorem 1.3 (Generalised separation) *Let X be Banach, $\Omega_1, \dots, \Omega_n$ be closed, $x_{0i} \in \Omega_i$ ($i = 1, \dots, n$), $\varepsilon > 0$, $\delta > 0$ and $\rho > 0$. Suppose that $\bigcap_{i=1}^n \Omega_i \cap (\rho \mathbb{B}_X) = \emptyset$, and $\|x_{0i}\| < \varepsilon$ ($i = 1, \dots, n$). The following assertions hold true:*

(i) *there exist $x_i \in \Omega_i \cap B_\delta(x_{0i})$, $x_i^* \in X^*$ ($i = 1, \dots, n$) and $v \in \rho \mathbb{B}_X$ such that*

$$\delta \sum_{i=1}^n d(x_i^*, N_{\Omega_i}(x_i)) + \rho \left\| \sum_{i=1}^n x_i^* \right\| < \varepsilon, \quad \sum_{i=1}^n \|x_i^*\| = 1, \quad (6)$$

$$\sum_{i=1}^n \langle x_i^*, v - x_i \rangle = \max_{1 \leq i \leq n} \|v - x_i\|;$$

(ii) *if X is Asplund, then, for any $\tau \in (0, 1)$, there exist $x_i \in \Omega_i \cap B_\delta(x_{0i})$, $x_i^* \in X^*$ ($i = 1, \dots, n$) and $v \in \rho \mathbb{B}_X$ such that condition (6) is satisfied, and*

$$\sum_{i=1}^n \langle x_i^*, v - x_i \rangle > \tau \max_{1 \leq i \leq n} \|v - x_i\|.$$

In view of our convention, Theorem 1.3 employs Clarke normal cones in part (i) and Fréchet normal cones in part (ii). Similar to the conventional extremal principle, it is a consequence of the Ekeland variational principle and corresponding subdifferential sum rules.

Remark 1.2 By [9, Corollary 3.2], the conclusions of Theorem 1.3 hold true under a weaker than $\|x_{0i}\| < \varepsilon$ ($i = 1, \dots, n$) condition

$$\max_{1 \leq i \leq n} \|x_{0i}\| < \inf_{u_i \in \Omega_i \ (i=1, \dots, n)} \inf_{u \in \rho \mathbb{B}_X} \max_{1 \leq i \leq n} \|u_i - u\| + \varepsilon.$$

The second (inner) infimum in the right-hand side of the inequality represents a convex minimisation problem of finding the *Chebyshev centre* of the set of points $\{u_1, \dots, u_n\}$; see, e.g., [2]. If an analytical solution of the problem can be found, the condition can be simplified; see [6].

2 Sequential extremality, stationarity and approximate stationarity

In this section, we refine the sequential extremality and stationarity concepts in Definition 1.2.

2.1 Relative sequential extremality, stationarity and approximate stationarity

We first discuss more specific than in Definition 1.2 extremality and stationarity properties of $\{\Omega_1, \dots, \Omega_n\}$ at a given sequence of points $\mathbf{x}^k := (x_1^k, \dots, x_n^k) \in \widehat{\Omega} := \Omega_1 \times \dots \times \Omega_n$ ($k \in \mathbb{N}$) with respect to a given sequence of translations $\mathbf{a}^k := (a_1^k, \dots, a_n^k) \in X^n \setminus \{0\}$ ($k \in \mathbb{N}$) such that $\mathbf{a}^k \rightarrow 0$ as $k \rightarrow +\infty$. We are going to use for these sequences the brief notations $\sigma_{\mathbf{x}} := \{\mathbf{x}^k\}$ and $\sigma_{\mathbf{a}} := \{\mathbf{a}^k\}$, respectively, and similar notations for some other sequences.

Given an $\mathbf{x} := (x_1, \dots, x_n) \in X^n$, denote $\zeta(\mathbf{x}) := d(0, \bigcap_{i=1}^n (\Omega_i - x_i))$.

The next definition introduces the key properties studied in the paper together with their notations which are used throughout the paper for brevity. Here and in what follows, **E**, **S** and **AS** stand for the extremality, stationarity and approximate stationarity, respectively.

Definition 2.1 (Relative sequential extremality, stationarity and approximate stationarity) The collection $\{\Omega_1, \dots, \Omega_n\}$ is

- (i) extremal at $\sigma_{\mathbf{x}}$ with respect to $\sigma_{\mathbf{a}}$ (denoted $\mathbf{E}(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}})$) if

$$\rho(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}}) := \liminf_{k \rightarrow +\infty} \zeta(\mathbf{x}^k + \mathbf{a}^k) > 0; \quad (7)$$

if $\rho(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}}) = +\infty$, we say that $\{\Omega_1, \dots, \Omega_n\}$ is globally extremal at $\sigma_{\mathbf{x}}$ with respect to $\sigma_{\mathbf{a}}$ (denoted $\mathbf{GE}(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}})$);

- (ii) α -stationary at $\sigma_{\mathbf{x}}$ with respect to $\sigma_{\mathbf{a}}$ (denoted $\mathbf{S}_{\alpha}(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}})$) if

$$\alpha(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}}) := \limsup_{k \rightarrow +\infty} \frac{\|\mathbf{a}^k\|}{\zeta(\mathbf{x}^k + \mathbf{a}^k)} < \alpha < +\infty; \quad (8)$$

if $\alpha(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}}) = 0$ (hence, (8) is satisfied with any $\alpha > 0$), we say that $\{\Omega_1, \dots, \Omega_n\}$ is stationary at $\sigma_{\mathbf{x}}$ with respect to $\sigma_{\mathbf{a}}$ (denoted $\mathbf{S}(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}})$);

- (iii) approximately α -stationary at $\sigma_{\mathbf{x}}$ with respect to $\sigma_{\mathbf{a}}$ (denoted $\mathbf{AS}_{\alpha}(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}})$) if

$$\tilde{\alpha}(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}}) := \inf_{\sigma_{\mathbf{u}} := \{\mathbf{u}^k\} \subset \hat{\Omega}, \mathbf{u}^k - \mathbf{x}^k \rightarrow 0} \alpha(\sigma_{\mathbf{u}}; \sigma_{\mathbf{a}}) < \alpha < +\infty; \quad (9)$$

if $\tilde{\alpha}(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}}) = 0$ (hence, (9) is satisfied with any $\alpha > 0$), we say that $\{\Omega_1, \dots, \Omega_n\}$ is approximately stationary at $\sigma_{\mathbf{x}}$ with respect to $\sigma_{\mathbf{a}}$ (denoted $\mathbf{AS}(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}})$).

The properties in Definition 2.1 are defined for an arbitrary sequence $\sigma_{\mathbf{x}} \subset \hat{\Omega}$, possibly unbounded. At the same time, the model studied in this paper covers the conventional ‘at-a-point’ setting. In the particular case when $x_i^k = \bar{x}$ for some $\bar{x} \in \bigcap_{i=1}^n \Omega_i$ and all $i = 1, \dots, n$ and $k = 1, 2, \dots$ (hence, $\zeta(\mathbf{x}^k + \mathbf{a}^k) = d(\bar{x}, \bigcap_{i=1}^n (\Omega_i - a_i))$), we write $\rho(\bar{x}; \sigma_{\mathbf{a}})$, $\alpha(\bar{x}; \sigma_{\mathbf{a}})$ and $\tilde{\alpha}(\bar{x}; \sigma_{\mathbf{a}})$ instead of $\rho(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}})$, $\alpha(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}})$ and $\tilde{\alpha}(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}})$, respectively, call the corresponding properties extremality, global extremality, α -stationarity, stationarity, approximate α -stationarity and approximate stationarity at $\bar{x}$ with respect to $\sigma_{\mathbf{a}}$, and employ the following respective notations: $\mathbf{E}(\bar{x}; \sigma_{\mathbf{a}})$, $\mathbf{GE}(\bar{x}; \sigma_{\mathbf{a}})$, $\mathbf{S}_{\alpha}(\bar{x}; \sigma_{\mathbf{a}})$, $\mathbf{S}(\bar{x}; \sigma_{\mathbf{a}})$, $\mathbf{AS}_{\alpha}(\bar{x}; \sigma_{\mathbf{a}})$ and $\mathbf{AS}(\bar{x}; \sigma_{\mathbf{a}})$.

Remark 2.1 (i) The numbers $\rho(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}})$, $\alpha(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}})$ and $\tilde{\alpha}(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}})$ computed in (7), (8) and (9) give exact quantitative characterisations of the respective properties, measuring their ‘power’.

- (ii) Our standing assumption that $\mathbf{a}^k \neq 0$ ($k \in \mathbb{N}$) ensures that the fraction in (8) is well-defined. In view of (7), in the case of extremality we automatically have $\mathbf{a}^k \neq 0$ for all sufficiently large $k \in \mathbb{N}$.
- (iii) The extremality property in part (i) of Definition 2.1 is fully determined by the sequence $\sigma_{\mathbf{y}} := \{\mathbf{y}^k\} \subset X^n$, where $\mathbf{y}^k := \mathbf{x}^k + \mathbf{a}^k$ ($k \in \mathbb{N}$) and does not depend on its representation as the sum of two sequences. So, it is possible to talk about extremality ‘with respect to $\sigma_{\mathbf{y}}$ ’. We adopt the terminology in the definition for consistency with that used in parts (ii) and (iii).
- (iv) Unlike the extremality, the stationarity properties in parts (ii) and (iii) of Definition 2.1 depend on both sequences $\sigma_{\mathbf{x}}$ and $\sigma_{\mathbf{a}}$. At the same time, the ‘ α -stationarity with respect to $\sigma_{\mathbf{y}}$ ’, defined by the condition

$$\alpha(\sigma_{\mathbf{y}}) := \limsup_{k \rightarrow +\infty} \frac{d(\mathbf{y}^k, \hat{\Omega})}{\zeta(\mathbf{y}^k)} < \alpha < +\infty$$

for a sequence $\sigma_{\mathbf{y}} := \{\mathbf{y}^k\} \subset X^n$, $\mathbf{y}^k := (y_1^k, \dots, y_n^k)$ satisfying $d(\mathbf{y}^k, \hat{\Omega}) > 0$ ($k \in \mathbb{N}$) and $d(\mathbf{y}^k, \hat{\Omega}) \rightarrow 0$ as $k \rightarrow +\infty$, can also be of interest. If $\mathbf{y}^k = \mathbf{x}^k + \mathbf{a}^k$, then obviously $\alpha(\sigma_{\mathbf{y}}) \leq$

$\alpha(\sigma_x; \sigma_a)$. Hence, this property is in general weaker than the one in part (ii) of Definition 2.1. Moreover, given a $\sigma_y \subset X^n$ as above, one can always find appropriate sequences $\sigma_x := \{\mathbf{x}^k\} \subset \widehat{\Omega}$ and $\sigma_a := \{\mathbf{a}^k\} \subset X^n$ such that $\mathbf{y}^k := \mathbf{x}^k + \mathbf{a}^k$ ($k \in \mathbb{N}$) and $\alpha(\sigma_y) = \alpha(\sigma_x; \sigma_a)$. For any $k \in \mathbb{N}$, we have

$$\zeta(\mathbf{y}^k) \geq \max_{1 \leq i \leq n} d(y_i^k, \Omega_i) = d(\mathbf{y}^k, \widehat{\Omega}).$$

It follows from the above estimate that $\alpha(\sigma_y) \leq 1$, and the property is automatically satisfied with any $\alpha > 1$. Hence, it only makes sense to study this property for small $\alpha > 0$.

- (v) The sequence σ_u in part (iii) of Definition 2.1 can be regarded as an approximation of the given sequence σ_x , justifying the name of the property.

The next two propositions are immediate consequences of Definition 2.1.

Proposition 2.1 (i) If $\rho(\sigma_x; \sigma_a) > 0$, then $\alpha(\sigma_x; \sigma_a) = 0$; (ii) $\tilde{\alpha}(\sigma_x; \sigma_a) \leq \alpha(\sigma_x; \sigma_a)$.

As a consequence, $\mathbf{GE}(\sigma_x; \sigma_a) \Rightarrow \mathbf{E}(\sigma_x; \sigma_a) \Rightarrow \mathbf{S}(\sigma_x; \sigma_a) \Rightarrow \mathbf{AS}(\sigma_x; \sigma_a)$, and

$\mathbf{S}_\alpha(\sigma_x; \sigma_a) \Rightarrow \mathbf{AS}_\alpha(\sigma_x; \sigma_a)$ for any $\alpha > 0$.

In particular, if $\bar{x} \in \bigcap_{i=1}^n \Omega_i$, then $\mathbf{GE}(\bar{x}; \sigma_a) \Rightarrow \mathbf{E}(\bar{x}; \sigma_a) \Rightarrow \mathbf{S}(\bar{x}; \sigma_a) \Rightarrow \mathbf{AS}(\bar{x}; \sigma_a)$, and $\mathbf{S}_\alpha(\bar{x}; \sigma_a) \Rightarrow \mathbf{AS}_\alpha(\bar{x}; \sigma_a)$ for any $\alpha > 0$.

Proposition 2.2 Let $\{\mathbf{x}^k, \mathbf{a}^k\}$ be a subsequence of $\{\mathbf{x}^k, \mathbf{a}^k\}$, $\sigma_{x'} := \{\mathbf{x}^k\}$ and $\sigma_{a'} := \{\mathbf{a}^k\}$. Then $\rho(\sigma_{x'}; \sigma_{a'}) \geq \rho(\sigma_x; \sigma_a)$, $\alpha(\sigma_{x'}; \sigma_{a'}) \leq \alpha(\sigma_x; \sigma_a)$ and $\tilde{\alpha}(\sigma_{x'}; \sigma_{a'}) \leq \tilde{\alpha}(\sigma_x; \sigma_a)$.

As a consequence, $\mathbf{P}(\sigma_x; \sigma_a) \Rightarrow \mathbf{P}(\sigma_{x'}; \sigma_{a'})$, where $\mathbf{P}$ stands for any of the properties: $\mathbf{E}$, $\mathbf{GE}$, $\mathbf{S}_\alpha$, $\mathbf{S}$, $\mathbf{AS}_\alpha$ or $\mathbf{AS}$. In particular, if $\bar{x} \in \bigcap_{i=1}^n \Omega_i$, then $\mathbf{P}(\bar{x}; \sigma_a) \Rightarrow \mathbf{P}(\bar{x}; \sigma_{a'})$.

We now illustrate the properties in Definition 2.1 by examples. The first one emphasises the role of the sequence σ_a of translations. We aim at providing some insights on how to choose appropriate translations ensuring the desired extremality or stationarity properties.

Example 2.1 Let $X := \mathbb{R}^2$, $\Omega_1 := \{(u, v) \in X \mid v \geq 0\}$, $\Omega_2 := \{(u, v) \in X \mid v \leq 0\}$ and $\bar{x} := (0, 0)$. The pair $\{\Omega_1, \Omega_2\}$ is obviously extremal at $\bar{x}$ in the sense of Definition 1.1(i): for any $\varepsilon > 0$, one can take $a_1 := (0, 0)$ and $a_2 := (0, \varepsilon/2)$.

We next show that the extremality and stationarity properties of this pair depend strongly on the choice of translations. Let $\mathbf{a}^k := (a_1^k, a_2^k) \in X^2 \setminus \{0\}$ and $\mathbf{a}^k \rightarrow 0$. Then $\Omega_1 - a_1^k = \{(u, v) \in \mathbb{R}^2 \mid v \geq -a_{12}^k\}$, $\Omega_2 - a_2^k = \{(u, v) \in \mathbb{R}^2 \mid v \leq -a_{22}^k\}$ and $A^k := (\Omega_1 - a_1^k) \cap (\Omega_2 - a_2^k) = \{(u, v) \in \mathbb{R}^2 \mid -a_{12}^k \leq v \leq -a_{22}^k\}$. Thus, $\zeta(\mathbf{a}^k) = +\infty$ if $a_{12}^k < a_{22}^k$, and $\zeta(\mathbf{a}^k) = \max\{-a_{12}^k, a_{22}^k, 0\} \leq \|\mathbf{a}^k\|$ otherwise. If $a_{12}^k < a_{22}^k$ for all sufficiently large $k \in \mathbb{N}$, then $\rho(\bar{x}; \sigma_a) = +\infty$. Otherwise, $\rho(\bar{x}; \sigma_a) = \liminf_{k \rightarrow +\infty} \zeta(\mathbf{a}^k) = 0$ and $\alpha(\bar{x}; \sigma_a) = \limsup_{k \rightarrow +\infty} \|\mathbf{a}^k\| / \zeta(\mathbf{a}^k) \geq 1$.

In the first case, $\{\Omega_1, \Omega_2\}$ is globally extremal at $\bar{x}$ with respect to σ_a . A typical example is given by $a_1^k := (0, 0)$, $a_2^k := (0, 1/k)$. This is basically the conventional extremality as in Definition 1.1(i).

In the second case, $\{\Omega_1, \Omega_2\}$ is neither extremal nor stationary at $\bar{x}$ with respect to σ_a . Moreover, $\{\Omega_1, \Omega_2\}$ is not approximately stationary at $\bar{x}$ with respect to σ_a . Indeed, if $\Omega_1 \ni u_1^k \rightarrow 0$, $\Omega_2 \ni u_2^k \rightarrow 0$ and $\mathbf{u}^k := (u_1^k, u_2^k)$, then $u_{12}^k \geq 0$ and $u_{22}^k \leq 0$; hence, $u_{12}^k + a_{12}^k \geq a_{12}^k$ and $u_{22}^k + a_{22}^k \leq a_{22}^k$. Similarly to the above,

$$\zeta(\mathbf{u}^k + \mathbf{a}^k) = \max\{-u_{12}^k - a_{12}^k, u_{22}^k + a_{22}^k, 0\} \leq \zeta(\mathbf{a}^k) \leq \|\mathbf{a}^k\|,$$

and consequently, $\alpha(\sigma_u; \sigma_a) \geq 1$, where $\sigma_u := \{\mathbf{u}^k\}$. Thus, $\tilde{\alpha}(\bar{x}; \sigma_a) \geq 1$. Hence, this case is of little interest from the point of view of applications.

The pair $\{\Omega_1, \Omega_2\}$ can be α -stationary at $\bar{x}$ with respect to σ_a with some $\alpha > 1$. For instance, if $a_1^k = a_2^k = (0, 1/k)$, then $\|\mathbf{a}^k\| = \zeta(\mathbf{a}^k) = 1/k$, and consequently, $\alpha(\bar{x}; \sigma_a) = 1$, i.e.,

$\{\Omega_1, \Omega_2\}$ is α -stationary at $\bar{x}$ with respect to σ_a with any $\alpha > 1$. On the other hand, if $a_1^k = (0, 1/k)$ and $a_2^k = (0, -1/k)$, then $\zeta(a^k) = 0$ and $\alpha(\bar{x}; \sigma_a) = +\infty$. Alternatively, one can take $a_1^k = (0, 1/k^{1/2})$ or $a_1^k = (1/k^{1/2}, 1/k)$, and $a_2^k = (0, 1/k)$. Then $\|a^k\| = 1/k^{1/2}$, $\zeta(a^k) = 1/k$, and consequently, $\alpha(\bar{x}; \sigma_a) = +\infty$. Thus, in these situations $\{\Omega_1, \Omega_2\}$ cannot be α -stationary at $\bar{x}$ with respect to σ_a with whatever $\alpha > 0$.

Note also that the first components of the vectors a_1^k and a_2^k play little or no role in this example. This is, of course, not surprising having in mind the geometry of the sets.

The next two examples illustrate differences between the properties in Definition 2.1; see Fig. 1.

Example 2.2 Let $X := \mathbb{R}^2$, $\Omega_1 := \{(u, v) \in X \mid u^2 + v \geq 0\}$, $\Omega_2 := \{(u, v) \in X \mid v \leq 0\}$ and $\bar{x} := (0, 0)$. Take $a_1^k := (0, 0)$, $a_2^k := (0, 1/k)$ and $a^k := (a_1^k, a_2^k)$. Then $\|a^k\| = 1/k$, $\Omega_1 - a_1^k = \Omega_1$, $\Omega_2 - a_2^k = \{(u, v) \in \mathbb{R}^2 \mid v \leq -1/k\}$, $A^k := (\Omega_1 - a_1^k) \cap (\Omega_2 - a_2^k) = \{(u, v) \in \mathbb{R}^2 \mid -u^2 \leq v \leq -1/k\}$ and $\zeta(a^k) = 1/k^{1/2}$. Hence, $\rho(\bar{x}; \sigma_a) = \alpha(\bar{x}; \sigma_a) = 0$, and consequently, $\{\Omega_1, \Omega_2\}$ is stationary but not extremal at $\bar{x}$ with respect to σ_a .

Example 2.3 Let $X := \mathbb{R}^2$, $\Omega_1 := \{(u, v) \in X \mid v \geq 0 \text{ or } u + v \geq 0\}$, $\Omega_2 := \{(u, v) \in X \mid v \leq 0\}$, and $\bar{x} := (0, 0)$. Take $a_1^k := (0, 0)$, $a_2^k := (0, 1/k^2)$ and $a^k := (a_1^k, a_2^k)$. Then $\|a^k\| = 1/k^2$, $\Omega_1 - a_1^k = \Omega_1$, $\Omega_2 - a_2^k = \{(u, v) \in \mathbb{R}^2 \mid v \leq -1/k^2\}$, $A^k := (\Omega_1 - a_1^k) \cap (\Omega_2 - a_2^k) = \{(u, v) \in \mathbb{R}^2 \mid -u \leq v \leq -1/k^2\}$ and $\zeta(a^k) = 1/k^2$. Hence, $\alpha(\bar{x}; \sigma_a) = 1$, and consequently, $\{\Omega_1, \Omega_2\}$ is not α -stationary at $\bar{x}$ with respect to σ_a with any $\alpha \in (0, 1]$.

Now, take $x_{12} = x_{21} = x_{22} = 0$, $x_{11} = -1/k$ and $x^k := (x_1^k, x_2^k)$. Then $\Omega_1 - x_1^k - a_1^k = \{(u, v) \in \mathbb{R}^2 \mid v \geq 0 \text{ or } u + v \geq 1/k\}$, $\Omega_2 - x_2^k - a_2^k = \{(u, v) \in \mathbb{R}^2 \mid v \leq -1/k^2\}$, $B^k := (\Omega_1 - x_1^k - a_1^k) \cap (\Omega_2 - x_2^k - a_2^k) = \{(u, v) \in \mathbb{R}^2 \mid 1/k - u \leq v \leq -1/k^2\}$ and $\zeta(x^k + a^k) = 1/k + 1/k^2$. Hence, $\alpha(\sigma_x; \sigma_a) = \lim_{k \rightarrow +\infty} \frac{1/k^2}{1/k + 1/k^2} = 0$, and consequently, $\{\Omega_1, \Omega_2\}$ is stationary at σ_x with respect to σ_a . By Definition 2.1(iii), $\tilde{\alpha}(\bar{x}; \sigma_a) = 0$, and consequently, $\{\Omega_1, \Omega_2\}$ is approximately stationary at $\bar{x}$ with respect to σ_a .

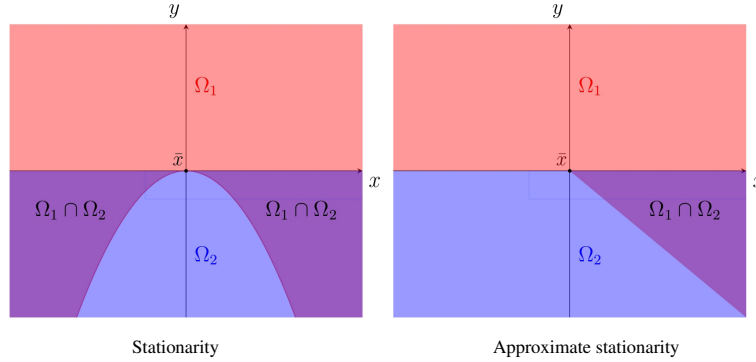


Fig. 1 Examples 2.2 and 2.3

The sequential extremality and approximate stationarity properties possess certain stability.

Proposition 2.3 Let $u^k \in \hat{\Omega}$ ($k \in \mathbb{N}$), $u^k - x^k \rightarrow 0$ as $k \rightarrow +\infty$ and $\sigma_u := \{u^k\}$. Then $\rho(\sigma_x; \sigma_a) = \rho(\sigma_u; \sigma_{a'})$ where $\sigma_{a'} := \{a'^k\}$, $a'^k := a^k + x^k - u^k$ ($k \in \mathbb{N}$), and $\tilde{\alpha}(\sigma_x; \sigma_a) = \tilde{\alpha}(\sigma_u; \sigma_a)$. As a consequence, $E(\sigma_x; \sigma_a) \Leftrightarrow E(\sigma_u; \sigma_{a'})$ and $AS(\sigma_x; \sigma_a) \Leftrightarrow AS(\sigma_u; \sigma_a)$.

In particular, if $\bar{x} \in \bigcap_{i=1}^n \Omega_i$, and $x_i^k \rightarrow \bar{x}$ as $k \rightarrow +\infty$ ($i = 1, \dots, n$), then $E(\bar{x}; \sigma_a) \Leftrightarrow E(\sigma_x; \sigma_{a'})$ where $a'^k := a^k + \bar{x} - x^k$ ($k \in \mathbb{N}$), and $AS(\bar{x}; \sigma_a) \Leftrightarrow AS(\sigma_x; \sigma_a)$.

Remark 2.2 (i) Under the assumptions of Proposition 2.3, if $\rho(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}}) > 0$, then $\mathbf{a}^k \neq 0$ for all sufficiently large $k \in \mathbb{N}$.

- (ii) The formulations of stability of the extremality and approximate stationarity properties in Proposition 2.3 are different. The approximate stationarity equality $\tilde{\alpha}(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}}) = \tilde{\alpha}(\sigma_{\mathbf{u}}; \sigma_{\mathbf{a}})$ involves different sequences $\sigma_{\mathbf{x}}$ and $\sigma_{\mathbf{u}}$ (such that $\mathbf{u}^k - \mathbf{x}^k \rightarrow 0$) and holds with the same sequence $\sigma_{\mathbf{a}}$, while to ensure the extremality equality $\rho(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}}) = \rho(\sigma_{\mathbf{u}}; \sigma_{\mathbf{a}'})$, we need a different sequence $\sigma_{\mathbf{a}'}$ (satisfying $\mathbf{a}'^k - \mathbf{a}^k \rightarrow 0$).
- (iii) In view of Example 2.3, the (non-approximate) stationarity lacks stability.

The next proposition gives characterisations of the properties in Definition 2.1 holding on some subsequence of the given sequence $\{\mathbf{x}^k, \mathbf{a}^k\}$.

Proposition 2.4 *Let $\{\mathbf{x}^k, \mathbf{a}^k\}$ be a subsequence of $\{\mathbf{x}^k, \mathbf{a}^k\}$, $\sigma_{\mathbf{x}'} := \{\mathbf{x}^k\}$ and $\sigma_{\mathbf{a}'} := \{\mathbf{a}^k\}$. Then*

- (i) $\mathbf{E}(\sigma_{\mathbf{x}'}; \sigma_{\mathbf{a}'}) \Leftrightarrow$ *there is a $\rho > 0$ such that, for any $\varepsilon > 0$, there exists an integer $k > \varepsilon^{-1}$ such that $\|\mathbf{a}^k\| < \varepsilon$ and $\zeta(\mathbf{x}^k + \mathbf{a}^k) > \rho$;*
- (ii) $\mathbf{GE}(\sigma_{\mathbf{x}'}; \sigma_{\mathbf{a}'}) \Leftrightarrow$ *the above condition holds for any $\rho > 0$;*
- (iii) $\mathbf{S}_{\alpha}(\sigma_{\mathbf{x}'}; \sigma_{\mathbf{a}'}) \Leftrightarrow$ *there is an $\alpha' \in (0, \alpha)$ such that, for any $\varepsilon > 0$, there exists an integer $k > \varepsilon^{-1}$ such that $\|\mathbf{a}^k\| < \varepsilon$ and $\|\mathbf{a}^k\| < \alpha' \zeta(\mathbf{x}^k + \mathbf{a}^k)$;*
- (iv) $\mathbf{S}(\sigma_{\mathbf{x}'}; \sigma_{\mathbf{a}'}) \Leftrightarrow$ *for any $\varepsilon > 0$, there exists an integer $k > \varepsilon^{-1}$ such that $\|\mathbf{a}^k\| < \varepsilon$ and $\|\mathbf{a}^k\| < \varepsilon \zeta(\mathbf{x}^k + \mathbf{a}^k)$;*
- (v) $\mathbf{AS}_{\alpha}(\sigma_{\mathbf{x}'}; \sigma_{\mathbf{a}'}) \Leftrightarrow$ *there is an $\alpha' \in (0, \alpha)$ such that, for any $\varepsilon > 0$, there exist an integer $k > \varepsilon^{-1}$ and points $x'_i \in \Omega_i \cap B_{\varepsilon}(x_i^k)$ ($i = 1, \dots, n$) such that $\|\mathbf{a}^k\| < \varepsilon$ and $\|\mathbf{a}^k\| < \alpha' \zeta(\mathbf{x}' + \mathbf{a}^k)$, where $\mathbf{x}' := (x'_1, \dots, x'_n)$;*
- (vi) $\mathbf{AS}(\sigma_{\mathbf{x}'}; \sigma_{\mathbf{a}'}) \Leftrightarrow$ *for any $\varepsilon > 0$, there exist an integer $k > \varepsilon^{-1}$ and points $x'_i \in \Omega_i \cap B_{\varepsilon}(x_i^k)$ ($i = 1, \dots, n$) such that $\|\mathbf{a}^k\| < \varepsilon$ and $\|\mathbf{a}^k\| < \varepsilon \zeta(\mathbf{x}' + \mathbf{a}^k)$, where $\mathbf{x}' := (x'_1, \dots, x'_n)$.*

All the assertions in Proposition 2.4 can be proved using basically the same arguments. Below, we provide a proof of assertion (i) as an illustration.

Proof (of Proposition 2.4(i)) Let $\mathbb{N} \ni k^j \rightarrow +\infty$ as $j \rightarrow +\infty$, $\{\Omega_1, \dots, \Omega_n\}$ be extremal at $\sigma_{\mathbf{x}'} := \{\mathbf{x}^{k^j}\}$ with respect to $\sigma_{\mathbf{a}'} := \{\mathbf{a}^{k^j}\}$, and let $\varepsilon > 0$. By Definition 2.1(i), $\rho(\sigma_{\mathbf{x}'}; \sigma_{\mathbf{a}'}) > 0$. Take any $\rho \in (0, \rho(\sigma_{\mathbf{x}'}; \sigma_{\mathbf{a}'}))$. By (7), there is a $j_0 \in \mathbb{N}$ such that $\zeta(\mathbf{x}^{k^j} + \mathbf{a}^{k^j}) > \rho$ for all integers $j > j_0$. Since $\mathbf{a}^k \rightarrow 0$, we can choose an integer $j > j_0$ so that $k := k^j > \varepsilon^{-1}$ and $\|\mathbf{a}^k\| < \varepsilon$. Condition $\zeta(\mathbf{x}^k + \mathbf{a}^k) > \rho$ is satisfied automatically.

Conversely, suppose that there is a $\rho > 0$ such that, for any $\varepsilon > 0$, there exists an integer $k > \varepsilon^{-1}$ such that $\|\mathbf{a}^k\| < \varepsilon$ and $\zeta(\mathbf{x}^k + \mathbf{a}^k) > \rho$. Let $j \in \mathbb{N}$. By the assumption, there exists an integer $k^j > j$ such that $\|\mathbf{a}^{k^j}\| < 1/j$ and $\zeta(\mathbf{x}^{k^j} + \mathbf{a}^{k^j}) > \rho$. Denote $\sigma_{\mathbf{x}'} := \{\mathbf{x}^{k^j}\}$ and $\sigma_{\mathbf{a}'} := \{\mathbf{a}^{k^j}\}$. Hence, $\rho(\sigma_{\mathbf{x}'}; \sigma_{\mathbf{a}'}) \geq \rho > 0$. By Definition 2.1(i), $\{\Omega_1, \dots, \Omega_n\}$ is extremal at $\sigma_{\mathbf{x}'}$ with respect to $\sigma_{\mathbf{a}'}$. $\square$

Corollary 2.1 *Let $\bar{x} \in \bigcap_{i=1}^n \Omega_i$, and $\sigma_{\mathbf{a}'}$ be a subsequence of $\sigma_{\mathbf{a}}$. Then*

- (i) $\mathbf{E}(\bar{x}; \sigma_{\mathbf{a}'}) \Leftrightarrow$ *there is a $\rho > 0$ such that, for any $\varepsilon > 0$, there exists an integer $k > \varepsilon^{-1}$ such that $\|\mathbf{a}^k\| < \varepsilon$ and $d(\bar{x}, \bigcap_{i=1}^n (\Omega_i - \mathbf{a}_i^k)) > \rho$;*
- (ii) $\mathbf{GE}(\bar{x}; \sigma_{\mathbf{a}'}) \Leftrightarrow$ *the above condition holds for any $\rho > 0$;*
- (iii) $\mathbf{S}_{\alpha}(\bar{x}; \sigma_{\mathbf{a}'}) \Leftrightarrow$ *there is an $\alpha' \in (0, \alpha)$ such that, for any $\varepsilon > 0$, there exists an integer $k > \varepsilon^{-1}$ such that $\|\mathbf{a}^k\| < \varepsilon$ and $\|\mathbf{a}^k\| < \alpha' d(\bar{x}, \bigcap_{i=1}^n (\Omega_i - \mathbf{a}_i^k))$;*
- (iv) $\mathbf{S}(\bar{x}; \sigma_{\mathbf{a}'}) \Leftrightarrow$ *for any $\varepsilon > 0$, there exists an integer $k > \varepsilon^{-1}$ such that $\|\mathbf{a}^k\| < \varepsilon$ and $\|\mathbf{a}^k\| < \varepsilon d(\bar{x}, \bigcap_{i=1}^n (\Omega_i - \mathbf{a}_i^k))$;*
- (v) $\mathbf{AS}_{\alpha}(\bar{x}; \sigma_{\mathbf{a}'}) \Leftrightarrow$ *for any $\varepsilon > 0$, there exist an integer $k > \varepsilon^{-1}$ and points $x'_i \in \Omega_i \cap B_{\varepsilon}(\bar{x})$ ($i = 1, \dots, n$) such that $\|\mathbf{a}^k\| < \varepsilon$ and $\|\mathbf{a}^k\| < \alpha' \zeta(\mathbf{x}' + \mathbf{a}^k)$, where $\mathbf{x}' := (x'_1, \dots, x'_n)$;*
- (vi) $\mathbf{AS}(\bar{x}; \sigma_{\mathbf{a}'}) \Leftrightarrow$ *for any $\varepsilon > 0$, there exist an integer $k > \varepsilon^{-1}$ and points $x'_i \in \Omega_i \cap B_{\varepsilon}(\bar{x})$ ($i = 1, \dots, n$) such that $\|\mathbf{a}^k\| < \varepsilon$ and $\|\mathbf{a}^k\| < \varepsilon \zeta(\mathbf{x}' + \mathbf{a}^k)$, where $\mathbf{x}' := (x'_1, \dots, x'_n)$.*

2.2 Sequential extremality, stationarity and approximate stationarity

In this subsection, we move closer to the sequential definitions studied in [10] as well as the conventional concepts of extremality and stationarity of a collection of sets and consider a more traditional setting when the extremality and stationarity properties of $\{\Omega_1, \dots, \Omega_n\}$ at a fixed sequence σ_x hold for *some* sequence σ_a . The definitions and statements below rely on the corresponding ones in Section 2.1.

Definition 2.2 The collection $\{\Omega_1, \dots, \Omega_n\}$ is extremal (resp., globally extremal, α -stationary, stationary, approximately α -stationary, approximately stationary) at σ_x if it is extremal (resp., globally extremal, α -stationary, stationary, approximately α -stationary, approximately stationary) at σ_x with respect to some $\sigma_a := \{a^k\} \subset X^n \setminus \{0\}$ ($k \in \mathbb{N}$) such that $a^k \rightarrow 0$ as $k \rightarrow +\infty$. The corresponding properties are denoted by $\mathbf{E}(\sigma_x)$, $\mathbf{GE}(\sigma_x)$, $\mathbf{S}_\alpha(\sigma_x)$, $\mathbf{S}(\sigma_x)$, $\mathbf{AS}_\alpha(\sigma_x)$ and $\mathbf{AS}(\sigma_x)$, respectively.

If $x_i^k = \bar{x}$ for some $\bar{x} \in \bigcap_{i=1}^n \Omega_i$ and all $i = 1, \dots, n$ and $k = 1, 2, \dots$, we say that $\{\Omega_1, \dots, \Omega_n\}$ is extremal (resp., globally extremal, α -stationary, stationary, approximately α -stationary, approximately stationary) at $\bar{x}$. The corresponding properties are denoted by $\mathbf{E}(\bar{x})$, $\mathbf{GE}(\bar{x})$, $\mathbf{S}_\alpha(\bar{x})$, $\mathbf{S}(\bar{x})$, $\mathbf{AS}_\alpha(\bar{x})$ or $\mathbf{AS}(\bar{x})$.

The next three statements are consequences of the corresponding ones in Section 2.1.

Proposition 2.5 $\mathbf{GE}(\sigma_x) \Rightarrow \mathbf{E}(\sigma_x) \Rightarrow \mathbf{S}(\sigma_x) \Rightarrow \mathbf{AS}(\sigma_x)$, and $\mathbf{S}_\alpha(\sigma_x) \Rightarrow \mathbf{AS}_\alpha(\sigma_x)$ for any $\alpha > 0$. In particular, if $\bar{x} \in \bigcap_{i=1}^n \Omega_i$, then

$\mathbf{GE}(\bar{x}) \Rightarrow \mathbf{E}(\bar{x}) \Rightarrow \mathbf{S}(\bar{x}) \Rightarrow \mathbf{AS}(\bar{x})$, and $\mathbf{S}_\alpha(\bar{x}) \Rightarrow \mathbf{AS}_\alpha(\bar{x})$ for any $\alpha > 0$.

Proposition 2.6 Let $\mathbf{P}$ stand for any of the properties: $\mathbf{E}$, $\mathbf{GE}$, $\mathbf{S}_\alpha$, $\mathbf{S}$, $\mathbf{AS}_\alpha$ or $\mathbf{AS}$.

Then $\mathbf{P}(\sigma_x) \Rightarrow \mathbf{P}(\sigma_{x'})$ for any subsequence $\sigma_{x'}$ of σ_x .

Proposition 2.7 Let $\sigma_u := \{u^k\} \subset \widehat{\Omega}$ and $u^k - x^k \rightarrow 0$ as $k \rightarrow +\infty$. Then $\mathbf{P}(\sigma_x) \Leftrightarrow \mathbf{P}(\sigma_u)$, where $\mathbf{P}$ stands for any of the properties: $\mathbf{E}$, $\mathbf{GE}$, $\mathbf{AS}_\alpha$ or $\mathbf{AS}$. In particular, if $x_i^k \rightarrow \bar{x}$ as $k \rightarrow +\infty$ for some $\bar{x} \in \bigcap_{i=1}^n \Omega_i$ and all $i = 1, \dots, n$, then $\mathbf{P}(\bar{x}) \Leftrightarrow \mathbf{P}(\sigma_x)$.

In view of Proposition 2.7, the sequential extremality and approximate stationarity properties in the sense of Definition 2.2 possess certain stability. The next modification of Example 2.3 shows that the (non-approximate) stationarity lacks stability.

Example 2.4 Let $X := \mathbb{R}^2$, $\Omega_1 := \{(u, v) \in X \mid v \geq 0 \text{ or } u + v \geq 0\}$, $\Omega_2 := \{(u, v) \in X \mid v \leq 0\}$, and $\bar{x} := (0, 0)$. It was shown in Example 2.3 that $\{\Omega_1, \Omega_2\}$ is stationary at some σ_x converging to $(\bar{x}, \bar{x})$ with respect to some σ_a converging to 0, hence, just stationary at σ_x , while it is not stationary at $\bar{x}$ with respect to that σ_a . It remains to check that $\{\Omega_1, \Omega_2\}$ is not stationary at $\bar{x}$ with respect to any σ_a converging to 0.

Let $a_1^k := (a_{11}^k, a_{12}^k)$, $a_2^k := (a_{21}^k, a_{22}^k)$, $a^k := (a_1^k, a_2^k)$, and $a^k \rightarrow 0$. Then $\Omega_1 - a_1^k = \{(u, v) \in \mathbb{R}^2 \mid v \geq -a_{12}^k \text{ or } u + v \geq -a_{11}^k - a_{12}^k\}$, $\Omega_2 - a_2^k = \{(u, v) \in \mathbb{R}^2 \mid v \leq -a_{22}^k\}$ and $A^k := (\Omega_1 - a_1^k) \cap (\Omega_2 - a_2^k) = \{(u, v) \in \mathbb{R}^2 \mid -\max\{u, a_{11}^k\} - a_{12}^k \leq v \leq -a_{22}^k\}$. If $a_{11}^k \geq a_{22}^k - a_{12}^k$, then $\zeta(a^k) = (a_{22}^k)_+$. Otherwise, $\zeta(a^k) = \max\{a_{22}^k, a_{22}^k - a_{12}^k, 0\}$. Thus, $\zeta(a^k) \leq 2\|a^k\|$. Hence, $\alpha(\bar{x}; \sigma_a) \geq 1/2$, and consequently, $\{\Omega_1, \Omega_2\}$ is not stationary at $\bar{x}$ with respect to σ_a .

Proposition 2.8 Let $\sigma_{x'}$ be a subsequence of σ_x . Then

- (i) $\mathbf{E}(\sigma_{x'}) \Leftrightarrow$ there is a $\rho > 0$ such that, for any $\varepsilon > 0$, there exist an integer $k > \varepsilon^{-1}$ and points $a_1, \dots, a_n \in \varepsilon \mathbb{B}_X$ such that $\zeta(x^k + a) > \rho$, where $a := (a_1, \dots, a_n)$;
- (ii) $\mathbf{GE}(\sigma_{x'}) \Leftrightarrow$ the above condition holds for any $\rho > 0$;
- (iii) $\mathbf{S}_\alpha(\sigma_{x'}) \Leftrightarrow$ there is an $\alpha' \in (0, \alpha)$ such that, for any $\varepsilon > 0$, there exist an integer $k > \varepsilon^{-1}$ and points $a_1, \dots, a_n \in \varepsilon \mathbb{B}_X$ such that $\|a\| < \alpha' \zeta(x^k + a)$, where $a := (a_1, \dots, a_n)$;

- (iv) $\mathbf{S}(\sigma_{\mathbf{x}}) \Leftrightarrow$ for any $\varepsilon > 0$, there exist an integer $k > \varepsilon^{-1}$ and points $a_1, \dots, a_n \in \varepsilon\mathbb{B}_X$ such that $\|\mathbf{a}\| < \varepsilon\zeta(\mathbf{x}^k + \mathbf{a})$, where $\mathbf{a} := (a_1, \dots, a_n)$;
- (v) $\mathbf{AS}_\alpha(\sigma_{\mathbf{x}'}) \Leftrightarrow$ there is an $\alpha' \in (0, \alpha)$ such that, for any $\varepsilon > 0$, there exist an integer $k > \varepsilon^{-1}$, and points $x'_i \in \Omega_i \cap B_\varepsilon(x_i^k)$ and $a_i \in \varepsilon\mathbb{B}_X$ ($i = 1, \dots, n$) such that $\|\mathbf{a}\| < \alpha'\zeta(\mathbf{x}' + \mathbf{a})$, where $\mathbf{x}' := (x'_1, \dots, x'_n)$ and $\mathbf{a} := (a_1, \dots, a_n)$;
- (vi) $\mathbf{AS}(\sigma_{\mathbf{x}'}) \Leftrightarrow$ for any $\varepsilon > 0$, there exist an integer $k > \varepsilon^{-1}$, and points $x'_i \in \Omega_i \cap B_\varepsilon(x_i^k)$ and $a_i \in \varepsilon\mathbb{B}_X$ ($i = 1, \dots, n$) such that $\|\mathbf{a}\| < \varepsilon\zeta(\mathbf{x}' + \mathbf{a})$, where $\mathbf{x}' := (x'_1, \dots, x'_n)$ and $\mathbf{a} := (a_1, \dots, a_n)$.

The proofs of all the assertions in the above proposition use standard arguments already demonstrated in the proof of Proposition 2.4(i). If $x_i^k = \bar{x}$ for some $\bar{x} \in \bigcap_{i=1}^n \Omega_i$ and all $i = 1, \dots, n$ and $k = 1, 2, \dots$, the statement takes a simpler form and does not involve subsequences.

Corollary 2.2 Let $\bar{x} \in \bigcap_{i=1}^n \Omega_i$.

- (i) $\mathbf{E}(\bar{x}) \Leftrightarrow$ there is a $\rho > 0$ such that, for any $\varepsilon > 0$, there exist $a_1, \dots, a_n \in \varepsilon\mathbb{B}_X$ such that $d(\bar{x}, \bigcap_{i=1}^n (\Omega_i - a_i)) > \rho$;
- (ii) $\mathbf{GE}(\bar{x}) \Leftrightarrow$ the above condition holds for any $\rho > 0$;
- (iii) $\mathbf{S}_\alpha(\bar{x}) \Leftrightarrow$ there is an $\alpha' \in (0, \alpha)$ such that, for any $\varepsilon > 0$, there exist points $a_1, \dots, a_n \in \varepsilon\mathbb{B}_X$ such that $\max_{1 \leq i \leq n} \|a_i\| < \alpha' d(\bar{x}, \bigcap_{i=1}^n (\Omega_i - a_i))$;
- (iv) $\mathbf{S}(\bar{x}) \Leftrightarrow$ for any $\varepsilon > 0$, there exist points $a_1, \dots, a_n \in \varepsilon\mathbb{B}_X$ such that $\max_{1 \leq i \leq n} \|a_i\| < \varepsilon d(\bar{x}, \bigcap_{i=1}^n (\Omega_i - a_i))$;
- (v) $\mathbf{AS}_\alpha(\bar{x}) \Leftrightarrow$ there is an $\alpha' \in (0, \alpha)$ such that, for any $\varepsilon > 0$, there exist points $x'_i \in \Omega_i \cap B_\varepsilon(\bar{x})$ and $a_i \in \varepsilon\mathbb{B}_X$ ($i = 1, \dots, n$) such that $\|\mathbf{a}\| < \alpha'\zeta(\mathbf{x}' + \mathbf{a})$, where $\mathbf{x}' := (x'_1, \dots, x'_n)$ and $\mathbf{a} := (a_1, \dots, a_n)$;
- (vi) $\mathbf{AS}(\bar{x}) \Leftrightarrow$ for any $\varepsilon > 0$, there exist points $x'_i \in \Omega_i \cap B_\varepsilon(\bar{x})$ and $a_i \in \varepsilon\mathbb{B}_X$ ($i = 1, \dots, n$) such that $\|\mathbf{a}\| < \varepsilon\zeta(\mathbf{x}' + \mathbf{a})$, where $\mathbf{x}' := (x'_1, \dots, x'_n)$ and $\mathbf{a} := (a_1, \dots, a_n)$.

Remark 2.3 In view of Proposition 2.8, in the particular case when $x_i^1 = x_i^2 = \dots$ ($i = 1, \dots, n$), properties $\mathbf{E}(\sigma_{\mathbf{x}})$, $\mathbf{S}(\sigma_{\mathbf{x}})$ and $\mathbf{AS}(\sigma_{\mathbf{x}})$ reduce to the corresponding ‘relative’ properties in [3, Definition 3] (studied there for $n = 2$). If all these points coincide (the case covered by Corollary 2.2), properties $\mathbf{E}(\bar{x})$, $\mathbf{S}(\bar{x})$ and $\mathbf{AS}(\bar{x})$ are equivalent to the corresponding extremality, stationarity and approximate stationarity properties in Definition 1.1.

In the convex case, the statement of Proposition 2.5 can be partially reversed.

Proposition 2.9 Let $\Omega_1, \dots, \Omega_n$ be convex. Then $\mathbf{GE}(\sigma_{\mathbf{x}}) \Leftrightarrow \mathbf{E}(\sigma_{\mathbf{x}}) \Leftrightarrow \mathbf{S}(\sigma_{\mathbf{x}}) \Leftrightarrow \mathbf{AS}(\sigma_{\mathbf{x}})$.

In particular, if $\bar{x} \in \bigcap_{i=1}^n \Omega_i$, then $\mathbf{GE}(\bar{x}) \Leftrightarrow \mathbf{E}(\bar{x}) \Leftrightarrow \mathbf{S}(\bar{x}) \Leftrightarrow \mathbf{AS}(\bar{x})$.

Proof Thanks to Proposition 2.5, we only need to show that $\mathbf{AS}(\sigma_{\mathbf{x}}) \Rightarrow \mathbf{GE}(\sigma_{\mathbf{x}})$. Suppose that $\{\Omega_1, \dots, \Omega_n\}$ is not globally extremal at $\sigma_{\mathbf{x}}$. By Proposition 2.8(ii), there exist numbers $\rho > 0$ and $\varepsilon > 0$ such that

$$d\left(0, \bigcap_{i=1}^n (\Omega_i - x_i^k - a_i)\right) \leq \rho \quad (10)$$

for all integers $k > \varepsilon^{-1}$ and all $a_1, \dots, a_n \in \varepsilon\mathbb{B}_X$. Fix a $\delta \in (0, \min\{\rho, \varepsilon/(1+\rho)\})$.

Let $\mathbf{u}^k := (u_1^k, \dots, u_n^k) \in \widehat{\Omega}$, $\mathbf{a}^k := (a_1^k, \dots, a_n^k) \in X^n \setminus \{0\}$, $\mathbf{u}^k - \mathbf{x}^k \rightarrow 0$ and $\mathbf{a}^k \rightarrow 0$ as $k \rightarrow +\infty$. Choose a $k_0 > \delta^{-1}$ ($> \varepsilon^{-1}$) so that $\|\mathbf{u}^k - \mathbf{x}^k\| < \delta$ and $\|\mathbf{a}^k\| < \delta^2$ for all integers $k > k_0$. Take any integer $k > k_0$, and set $t := \|\mathbf{a}^k\|/(\delta\rho)$. Then $0 < t < \delta/\rho < 1$ and $\|\mathbf{u}^k - \mathbf{x}^k + \mathbf{a}^k/t\| < \delta(1+\rho) < \varepsilon$. By (10), $\zeta(\mathbf{u}^k + \mathbf{a}^k/t) \leq \rho$, and consequently, there is an $x \in \rho\mathbb{B}_X$ such that $x + u_i^k + a_i^k/t \in \Omega_i$ ($i = 1, \dots, n$). Since the sets are convex, we have $t(x + u_i^k + a_i^k/t) + (1-t)u_i^k \in \Omega_i$, or equivalently, $tx + u_i^k + a_i^k \in \Omega_i$ ($i = 1, \dots, n$), i.e., $tx \in \bigcap_{i=1}^n (\Omega_i - u_i^k - a_i^k)$. Hence, $\zeta(\mathbf{u}^k + \mathbf{a}^k) \leq t\|\mathbf{x}\| \leq t\rho = \|\mathbf{a}^k\|/\delta$, and consequently, $\|\mathbf{a}^k\|/\zeta(\mathbf{u}^k + \mathbf{a}^k) \geq \delta$ for all $k > k_0$. By Definitions 2.1 and 2.2, $\{\Omega_1, \dots, \Omega_n\}$ is not approximately stationary at $\sigma_{\mathbf{x}}$. $\square$

Remark 2.4 As discussed in the Introduction, the sequential Definition 2.2 (together with Definition 2.1 which it builds upon) partially extend and improve [10, Definitions 2.1 and 3.1] along several lines.

- (i) The model adopted in the current paper does not assume condition (1) as it is superfluous when proving dual necessary conditions.
- (ii) Ignoring the superfluous condition (1), thanks to Proposition 2.8, extremality and stationarity at $\{x_i^k\}$ ($i = 1, \dots, n$) in the sense of Definition 1.2 are equivalent to the corresponding properties in the sense of Definition 2.2 at some subsequence of σ_x . Thus, the definitions adopted in this paper are more specific in identifying the sequences relevant for the properties.
- (iii) The definition of approximate stationarity in part (iii) of Definition 2.1 differs from that in Definition 1.2(iii). Nevertheless, they are largely equivalent up to taking subsequences (ignoring the superfluous condition (1)).

In view of the above observations, Propositions 2.7 and 2.9 extend [10, Proposition 2.4(i) and (iii), and Proposition 2.7], respectively. Example 2.4 shows, in particular, that the claim in [10, Proposition 2.4(ii)] is not entirely correct. Many examples provided in [10] are applicable for illustrating the properties in Definition 2.2.

3 Generalised separation

In this section, we establish dual necessary (generalised separation) conditions for the sequential extremality and stationarity properties discussed in Section 2. The underlying space X is assumed here to be Banach, and the sets $\Omega_1, \dots, \Omega_n$ closed (unless stated otherwise).

The dual conditions are formulated in terms of normal cones to the sets involved. Recall our standing convention that the normal cone generic notation N stands for N^C if X is a general Banach space, and N^F if X is Asplund. Thus, the definition of the $(\mathbf{GS}_\alpha)$ property below covers both cases.

In the statements, we employ certain generalised separation conditions. Let $x_i^k \in \Omega_i$, $x_i^{*k} \in N_{\Omega_i}(x_i^k)$ ($i = 1, \dots, n$, $k = 1, 2, \dots$), and $\alpha > 0$.

(GS $_\alpha$) *Generalised α -separation*: $\limsup_{k \rightarrow +\infty} \|\sum_{i=1}^n x_i^{*k}\| < \alpha$ and $\sum_{i=1}^n \|x_i^{*k}\| = 1$ ($k \in \mathbb{N}$).

If $\lim_{k \rightarrow +\infty} \sum_{i=1}^n x_i^{*k} = 0$, we talk about the *generalised separation* and write **(GS)**.

The generalised separation conditions can be complemented by a certain auxiliary primal-dual condition connecting the dual variables (normals) in the generalised separation conditions and the corresponding primal variables. Let $\mathbf{z}^k \in X^n$, $\mathbf{x}^{*k} \in (X^*)^n$ and $\|\mathbf{x}^{*k}\| = 1$ ($k \in \mathbb{N}$).

(PD) *Complimentary primal-dual condition*: $\mathbf{z}^k \neq 0$ for all large $k \in \mathbb{N}$, and $\lim_{k \rightarrow +\infty} \frac{\langle \mathbf{x}^{*k}, \mathbf{z}^k \rangle}{\|\mathbf{z}^k\|} = 1$.

Remark 3.1 Condition **(PD)** roughly says that the primal unit vector $\mathbf{z}^k / \|\mathbf{z}^k\|$ and dual unit vector $\mathbf{x}^{*k}$ are getting almost ‘co-linear’ as $k \rightarrow +\infty$. More precisely, it is not difficult to show that, if X is Hilbert, then $\|\frac{\mathbf{z}^k}{\|\mathbf{z}^k\|} - \mathbf{x}^{*k}\| \rightarrow 0$ as $k \rightarrow +\infty$.

Theorem 3.1 *Let $\alpha > 0$. Suppose that $\{\Omega_1, \dots, \Omega_n\}$ is approximately α -stationary (particularly, α -stationary) at $\sigma_{\mathbf{x}_0} := \{\mathbf{x}_0^k\} \subset \widehat{\Omega}$. Then there exist $\mathbf{x}^k := (x_1^k, \dots, x_n^k)$ and $x_i^{*k} \in N_{\Omega_i}(x_i^k)$ ($i = 1, \dots, n$, $k \in \mathbb{N}$) such that $\mathbf{x}^k - \mathbf{x}_0^k \rightarrow 0$ as $k \rightarrow +\infty$, and condition **(GS $_\alpha$)** holds true.*

*Furthermore, if the assumed property holds with respect to some $\sigma_{\mathbf{a}} := \{\mathbf{a}^k\} \subset X^n \setminus \{0\}$, $\mathbf{a}^k := (a_1^k, \dots, a_n^k)$ ($k \in \mathbb{N}$) such that $\mathbf{a}^k \rightarrow 0$ as $k \rightarrow +\infty$, then there exist $\mathbf{u}^k := (u_1^k, \dots, u_n^k) \in \widehat{\Omega}$ and $v^k \in X$ ($k \in \mathbb{N}$) such that $\mathbf{u}^k - \mathbf{x}_0^k \rightarrow 0$ ($\mathbf{u}^k = \mathbf{x}_0^k$ if $\{\Omega_1, \dots, \Omega_n\}$ is α -stationary at $\sigma_{\mathbf{x}_0}$ with respect to $\sigma_{\mathbf{a}}$), $v^k \rightarrow 0$ as $k \rightarrow +\infty$, and condition **(PD)** is satisfied with $\mathbf{x}^{*k} := (x_1^{*k}, \dots, x_n^{*k})$,*

$$z_i^k := v^k - x_i^k + u_i^k + a_i^k \quad (i = 1, \dots, n) \quad \text{and} \quad \mathbf{z}^k := (z_1^k, \dots, z_n^k) \quad (k \in \mathbb{N}). \quad (11)$$

Proof By Definition 2.2, $\{\Omega_1, \dots, \Omega_n\}$ is approximately α -stationary at $\sigma_{\mathbf{x}_0}$ with respect to some $\sigma_{\mathbf{a}} := \{\mathbf{a}^k\} \subset X^n \setminus \{0\}$, $\mathbf{a}^k := (a_1^k, \dots, a_n^k)$ ($k \in \mathbb{N}$) such that $\mathbf{a}^k \rightarrow 0$ as $k \rightarrow +\infty$. By Definition 2.1(iii), there exist an $\alpha' \in (0, \alpha)$, a sequence $\mathbf{u}^k := (u_1^k, \dots, u_n^k) \in \widehat{\Omega}$ ($k \in \mathbb{N}$), and a $k_0 \in \mathbb{N}$ such that $\mathbf{u}^k - \mathbf{x}_0^k \rightarrow 0$ as $k \rightarrow +\infty$ and, for all integers $k > k_0$, it holds $d(0, \bigcap_{i=1}^n \Omega_i^k) > \rho^k$, where $\Omega_i^k := \Omega_i - u_i^k - a_i^k$ and $\rho^k := \|\mathbf{a}^k\|/\alpha'$. If $\{\Omega_1, \dots, \Omega_n\}$ is α -stationary at $\sigma_{\mathbf{x}_0}$ with respect to $\sigma_{\mathbf{a}}$, then, in view of Definition 2.1(ii), $\mathbf{u}^k = \mathbf{x}_0^k$ ($k \in \mathbb{N}$). Hence,

$$\bigcap_{i=1}^n \Omega_i^k \cap (\rho^k \mathbb{B}_X) = \emptyset. \quad (12)$$

Denote $t^k := \|\mathbf{a}^k\|$. Taking a larger k_0 if necessary, we assume without loss of generality that $(1+t^k)(t^k)^{\frac{1}{2}} < 1$ for all integers $k > k_0$.

Fix an integer $k > k_0$. Choose an $\alpha'' \in (\alpha', \alpha)$. Set $\varepsilon^k := t^k + \min\{(t^k)^2, \rho^k(\alpha'' - \alpha')\}$ and $\delta^k := (t^k)^{\frac{1}{2}}$. Then $\varepsilon^k > t^k$,

$$\limsup_{k \rightarrow +\infty} \frac{\varepsilon^k}{\rho^k} \leq \alpha' + (\alpha'' - \alpha') = \alpha'' < \alpha \quad \text{and} \quad \limsup_{k \rightarrow +\infty} \frac{\varepsilon^k}{\delta^k} \leq \lim_{k \rightarrow +\infty} (1+t^k)(t^k)^{\frac{1}{2}} = 0. \quad (13)$$

Note that $-a_i^k \in \Omega_i^k$ and $\|a_i^k\| < \varepsilon^k$ ($i = 1, \dots, n$). By Theorem 1.3(i), there exist $x_i^k \in \Omega_i$, $x_i'^k \in N_{\Omega_i}^C(x_i^k)$, $y_i^k \in X^*$ ($i = 1, \dots, n$) and $v^k \in \rho^k \mathbb{B}_X$ such that

$$\|x_i^k - u_i^k\| < \delta^k \quad (i = 1, \dots, n), \quad (14)$$

$$\sum_{i=1}^n \|y_i^k\| = 1, \quad (15)$$

$$\delta^k \sum_{i=1}^n \|x_i'^k - y_i^k\| + \rho^k \left\| \sum_{i=1}^n y_i^k \right\| < \varepsilon^k, \quad (16)$$

$$\sum_{i=1}^n \langle y_i^k, z_i^k \rangle = \max_{1 \leq i \leq n} \|z_i^k\|, \quad (17)$$

where z_i^k ($i = 1, \dots, n$) are given by (11). Moreover, $\mathbf{z}^k := (z_1^k, \dots, z_n^k) \neq 0$ because otherwise $v^k \in \bigcap_{i=1}^n \Omega_i^k \cap (\rho^k \mathbb{B}_X)$ contradicting (12).

Thus, $x_i^k - x_{0i}^k \rightarrow 0$ ($i = 1, \dots, n$), and $v^k \rightarrow 0$ as $k \rightarrow +\infty$. By (16),

$$\sum_{i=1}^n \|x_i'^k - y_i^k\| < \frac{\varepsilon^k}{\delta^k} < 1. \quad (18)$$

It follows from (15) and (18) that

$$1 - \frac{\varepsilon^k}{\delta^k} < \sum_{i=1}^n \|x_i'^k\| < 1 + \frac{\varepsilon^k}{\delta^k}$$

and, in view of (13),

$$0 \neq \eta^k := \sum_{i=1}^n \|x_i'^k\| \rightarrow 1 \quad \text{as } k \rightarrow +\infty. \quad (19)$$

Set $x_i^{*k} := x_i'^k / \eta^k$ ($i = 1, \dots, n$) and $\mathbf{x}^{*k} := (x_1^{*k}, \dots, x_n^{*k})$. Thus, $x_i^{*k} \in N_{\Omega_i}^C(x_i^k)$ ($i = 1, \dots, n$), $\|\mathbf{x}^{*k}\| = 1$,

$$\begin{aligned} \eta^k \left\| \sum_{i=1}^n x_i^{*k} \right\| &\leq \left\| \sum_{i=1}^n y_i^k \right\| + \sum_{i=1}^n \|x_i'^k - y_i^k\| \stackrel{(16), (18)}{<} \frac{\varepsilon^k}{\rho^k} + \frac{\varepsilon^k}{\delta^k}, \\ \eta^k \frac{\langle \mathbf{x}^{*k}, \mathbf{z}^k \rangle}{\|\mathbf{z}^k\|} &> \frac{\sum_{i=1}^n \langle y_i^k, z_i^k \rangle}{\|\mathbf{z}^k\|} - \sum_{i=1}^n \|x_i'^k - y_i^k\| \stackrel{(17), (18)}{>} 1 - \frac{\varepsilon^k}{\delta^k}. \end{aligned}$$

Taking limits in the above estimates, in view of (19) and (13), we obtain

$$\limsup_{k \rightarrow +\infty} \left\| \sum_{i=1}^n x_i^{*k} \right\| < \alpha \quad \text{and} \quad \lim_{k \rightarrow +\infty} \frac{\langle \mathbf{x}^{*k}, \mathbf{z}^k \rangle}{\|\mathbf{z}^k\|} = 1. \quad (20)$$

Let X be Asplund. By Theorem 1.3(ii), there exist $x_i^k \in \Omega_i$, $x_i^{*k} \in N_{\Omega_i}^F(x_i^k)$, $y_i^{*k} \in X^*$ ($i = 1, \dots, n$) and $v^k \in \rho^k \mathbb{B}_X$ such that conditions (14), (15) and (16) are satisfied, and

$$\sum_{i=1}^n \langle y_i^{*k}, z_i^k \rangle > \left(1 - \frac{1}{k}\right) \max_{1 \leq i \leq n} \|z_i^k\|, \quad (21)$$

where z_i^k ($i = 1, \dots, n$) are given by (11). As above, $\mathbf{z}^k := (z_1^k, \dots, z_n^k) \neq 0$, $x_i^k - x_{0i}^k \rightarrow 0$ ($i = 1, \dots, n$), $\eta^k := \sum_{i=1}^n \|x_i^{*k}\| \rightarrow 1$ as $k \rightarrow +\infty$, $x_i^{*k} := x_i^{*k}/\eta^k \in N_{\Omega_i}^F(x_i^k)$ ($i = 1, \dots, n$), and conditions (20) are satisfied with $\mathbf{x}^{*k} := (x_1^{*k}, \dots, x_n^{*k})$.

To formally complete the proof of condition $(\mathbf{GS}_\alpha)$, it suffices to set $x_i^k := x_i^{k_0+1}$, $x_i^{*k} := x_i^{(k_0+1)*}$ ($i = 1, \dots, n$) and $\mathbf{x}^{*k} := \mathbf{x}^{(k_0+1)*}$ for all $k = 1, \dots, k_0$. $\square$

The approximate stationarity property (i.e., the case $\tilde{\alpha}(\sigma_{\mathbf{x}_0}; \sigma_{\mathbf{a}}) = 0$) requires some additional effort.

Corollary 3.1 *Let $\sigma_{\mathbf{x}_0} := \{\mathbf{x}_0^k\} \subset \widehat{\Omega}$, $\mathbf{x}_0^k := (x_{01}^k, \dots, x_{0n}^k)$ ($k \in \mathbb{N}$). Suppose that $\{\Omega_1, \dots, \Omega_n\}$ is approximately stationary (particularly, stationary or extremal) at $\sigma_{\mathbf{x}_0}$. Then there exist $x_i^k \in \Omega_i$, $x_i^{*k} \in N_{\Omega_i}(x_i^k)$ ($i = 1, \dots, n$, $k = 1, 2, \dots$) such that $x_i^k - x_{0i}^k \rightarrow 0$ ($i = 1, \dots, n$) as $k \rightarrow +\infty$, and condition $(\mathbf{GS})$ holds true.*

*Furthermore, if the assumed property holds with respect to some $\sigma_{\mathbf{a}} := \{\mathbf{a}^k\} \subset X^n \setminus \{0\}$, $\mathbf{a}^k := (a_1^k, \dots, a_n^k)$ ($k \in \mathbb{N}$) such that $\mathbf{a}^k \rightarrow 0$ as $k \rightarrow +\infty$, then there exist $\mathbf{u}^k := (u_1^k, \dots, u_n^k) \in \widehat{\Omega}$ and $v^k \in X$ ($k \in \mathbb{N}$) such that $\mathbf{u}^k - \mathbf{x}_0^k \rightarrow 0$ ($\mathbf{u}^k = \mathbf{x}_0^k$ if $\{\Omega_1, \dots, \Omega_n\}$ is stationary or extremal at $\sigma_{\mathbf{x}_0}$ with respect to $\sigma_{\mathbf{a}}$), $v^k \rightarrow 0$ as $k \rightarrow +\infty$, and condition $(\mathbf{PD})$ is satisfied with $\mathbf{x}^{*k} := (x_1^{*k}, \dots, x_n^{*k})$ and $\mathbf{z}^k$ given by (11).*

Proof By Definitions 2.2 and 2.1(iii), $\{\Omega_1, \dots, \Omega_n\}$ is approximately α -stationary at $\sigma_{\mathbf{x}_0}$ with respect to some $\sigma_{\mathbf{a}} := \{\mathbf{a}^k\} \subset X^n \setminus \{0\}$, $\mathbf{a}^k := (a_1^k, \dots, a_n^k)$ ($k \in \mathbb{N}$) such that $\mathbf{a}^k \rightarrow 0$ as $k \rightarrow +\infty$ for any $\alpha > 0$. Let $j \in \mathbb{N}$. By Theorem 3.1, there exist $x_i^k(j), u_i^k(j) \in \Omega_i$, $x_i^{*k}(j) \in N_{\Omega_i}(x_i^k(j))$ ($i = 1, \dots, n$) and $v^k(j) \in X$ ($k \in \mathbb{N}$) such that $x_i^k(j) - x_{0i}^k \rightarrow 0$, $u_i^k(j) - x_{0i}^k \rightarrow 0$, $v^k(j) \rightarrow 0$, $\limsup_{k \rightarrow +\infty} \|\sum_{i=1}^n x_i^{*k}(j)\| \leq 1/j$, $\sum_{i=1}^n \|x_i^{*k}(j)\| = 1$, $\mathbf{z}^k(j) := (z_1^k(j), \dots, z_n^k(j)) \neq 0$ for all sufficiently large $k \in \mathbb{N}$, and $\lim_{k \rightarrow +\infty} \langle \mathbf{x}^{*k}(j), \mathbf{z}^k(j) \rangle / \|\mathbf{z}^k(j)\| = 1$, where $z_i^k(j) := v^k(j) - x_i^k(j) + u_i^k(j) + a_i^k$ ($i = 1, \dots, n$) and $\mathbf{x}^{*k}(j) := (x_1^{*k}(j), \dots, x_n^{*k}(j))$. If $\{\Omega_1, \dots, \Omega_n\}$ is stationary or extremal at $\sigma_{\mathbf{x}_0}$ with respect to $\sigma_{\mathbf{a}}$, then, in view of Definition 2.1(ii), $u_i^k(j) = x_{0i}^k(j)$ ($i = 1, \dots, n$, $k = 1, 2, \dots$). Hence, there is an integer $k^j > j$ such that $\|\sum_{i=1}^n x_i^{*k}(j)\| < 2/j$ for all integers $k \geq k^j$. Note that $k^j \rightarrow +\infty$ as $j \rightarrow +\infty$.

For each $k \in \mathbb{N}$, set $j^k := \max\{j \in \mathbb{N} \mid k^j \leq k\}$ if $k \geq k^1$ and $j^k := 1$ otherwise. Thus, $j^k \rightarrow +\infty$ as $k \rightarrow +\infty$. Setting $x_i^k := x_i^k(j^k)$, $u_i^k := u_i^k(j^k)$, $x_i^{*k} := x_i^{*k}(j^k)$ ($i = 1, \dots, n$), $v^k := v^k(j^k)$ and $\mathbf{z}^k := \mathbf{z}^k(j^k)$, we see that $x_i^k - x_{0i}^k \rightarrow 0$, $u_i^k - x_{0i}^k \rightarrow 0$ ($i = 1, \dots, n$), $v^k \rightarrow 0$, and $\sum_{i=1}^n x_i^{*k} \rightarrow 0$ as $k \rightarrow +\infty$. Hence, conditions $(\mathbf{GS})$ and $(\mathbf{PD})$ are satisfied. $\square$

Remark 3.2 (i) The generalised separation conditions $(\mathbf{GS}_\alpha)$ and $(\mathbf{GS})$ are the main dual necessary conditions in Theorem 3.1 and Corollary 3.1. The complementary primal-dual condition $(\mathbf{PD})$ provides additional restrictions on the dual variables.

(ii) In view of Remark 2.4, Theorem 3.1 and Corollary 3.1 improve [10, Theorem 3.2 and Corollaries 3.3 and 3.4].

- (iii) In the particular case when $x_{0i}^k = \bar{x}$ for some $\bar{x} \in \bigcap_{i=1}^n \Omega_i$ and all $i = 1, \dots, n$ and $k = 1, 2, \dots$, Theorem 3.1 and Corollary 3.1 give dual necessary conditions for the specified properties at $\bar{x}$. Conditions $x_i^k - x_{0i}^k \rightarrow 0$ and $u_i^k - x_{0i}^k \rightarrow 0$ reduce in this case to $x_i^k \rightarrow \bar{x}$ and $u_i^k \rightarrow \bar{x}$, respectively. With this in mind, Corollary 3.1 extends the conventional extremal principle [24,27,30] and its numerous generalisations.
- (iv) Imposing certain sequential normal compactness assumptions (which are automatically satisfied in finite dimensions), one can formulate limiting versions of Theorem 3.1 and Corollary 3.1 in terms of certain types of limiting normal cones. This topic is going to be explored in more detail elsewhere.

The assertions in the first parts of Theorem 3.1 and Corollary 3.1 can be partially reversed, i.e., the dual necessary generalised separation conditions discussed in this section are in a sense also sufficient. To show this, we need to prove first the next lemma which is of independent interest.

Lemma 3.1 *Let $\mathbf{x} := (x_1, \dots, x_n) \in \widehat{\Omega}$, $x_i^* \in N_{\Omega_i}^F(x_i)$ ($i = 1, \dots, n$), $\|\sum_{i=1}^n x_i^*\| < \alpha$, $\sum_{i=1}^n \|x_i^*\| = 1$, and $\varepsilon > 0$. Then there exists an $\mathbf{a} := (a_1, \dots, a_n) \in \varepsilon \mathbb{B}_{X^n} \setminus \{0\}$ such that $\|\mathbf{a}\|/\zeta(\mathbf{x} + \mathbf{a}) < \alpha$.*

Proof Observe that $\mathbf{x}^* := (x_1^*, \dots, x_n^*) \in N_{\widehat{\Omega}}^F(\mathbf{x})$ (see, e.g., [9, Proposition 2.6]). Choose an $\alpha' \in (\|\sum_{i=1}^n x_i^*\|, \alpha)$ and set $\xi := (\alpha - \alpha')/2$. By the definition of Fréchet normal cone, there is a $\rho \in (0, \varepsilon/\alpha)$ such that

$$\langle \mathbf{x}^*, \mathbf{u} - \mathbf{x} \rangle \leq \frac{\xi}{\alpha + 1} \|\mathbf{u} - \mathbf{x}\| < \xi \rho \quad \text{for all } \mathbf{u} \in \widehat{\Omega} \cap B_{(\alpha+1)\rho}(\mathbf{x}). \quad (22)$$

Choose a nonzero $\mathbf{a} := (a_1, \dots, a_n) \in \alpha \rho \mathbb{B}_{X^n} (\subset \varepsilon \mathbb{B}_{X^n})$ such that

$$\langle \mathbf{x}^*, \mathbf{a} \rangle > (\alpha - \xi) \rho = (\alpha' + \xi) \rho. \quad (23)$$

Suppose that $\zeta(\mathbf{x} + \mathbf{a}) < \rho$. Then $u_i - x_i - a_i = x_0$ for some $x_0 \in \rho \mathbb{B}_X$ and $\mathbf{u} := (u_1, \dots, u_n) \in \widehat{\Omega}$, and all $i = 1, \dots, n$. Hence, $\|\mathbf{u} - \mathbf{x}\| \leq \|\mathbf{a}\| + \|x_0\| < (\alpha + 1)\rho$, and, by (22), $\langle \mathbf{x}^*, \mathbf{u} - \mathbf{x} \rangle < \xi \rho$. Employing (23), we obtain

$$-\alpha' \rho < -\left\| \sum_{i=1}^n x_i^* \right\| \|x_0\| \leq \left\langle \sum_{i=1}^n x_i^*, x_0 \right\rangle = \langle \mathbf{x}^*, \mathbf{u} - \mathbf{x} \rangle - \langle \mathbf{x}^*, \mathbf{a} \rangle < -\alpha' \rho.$$

The contradiction proves that $\zeta(\mathbf{x} + \mathbf{a}) \geq \rho$. Hence, $\|\mathbf{a}\|/\zeta(\mathbf{x} + \mathbf{a}) < \alpha$. $\square$

As a consequence of Lemma 3.1, we have the promised statement partially reversing the assertions in the first parts of Theorem 3.1 and Corollary 3.1.

Proposition 3.1 *Let $\sigma_{\mathbf{x}} := \{\mathbf{x}^k\} \subset \widehat{\Omega}$, $\mathbf{x}^k := (x_1^k, \dots, x_n^k)$, and $x_i^{*k} \in N_{\Omega_i}^F(x_i^k)$ ($i = 1, \dots, n$, $k = 1, 2, \dots$). Then $(\mathbf{GS}) \Rightarrow \mathbf{S}(\sigma_{\mathbf{x}})$ and $(\mathbf{GS}_{\alpha}) \Rightarrow \mathbf{S}_{\alpha}(\sigma_{\mathbf{x}})$ for any $\alpha > 0$.*

Proof Let $\varepsilon^k > 0$ ($k \in \mathbb{N}$), $\varepsilon^k \downarrow 0$ as $k \rightarrow +\infty$, and condition $(\mathbf{GS}_{\alpha})$ be satisfied. Choose an $\alpha' < \alpha$ so that $\|\sum_{i=1}^n x_i^{*k}\| < \alpha'$ for all sufficiently large $k \in \mathbb{N}$. By Lemma 3.1, for each such k , there exists an $\mathbf{a}^k := (a_1^k, \dots, a_n^k) \in \varepsilon^k \mathbb{B}_{X^n} \setminus \{0\}$ such that $\|\mathbf{a}^k\|/\zeta(\mathbf{x}^k + \mathbf{a}^k) < \alpha'$. Hence, $\mathbf{a}^k \rightarrow 0$ and $\alpha(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}}) \leq \alpha' < \alpha$, where $\sigma_{\mathbf{a}} := \{\mathbf{a}^k\}$, i.e., $\{\Omega_1, \dots, \Omega_n\}$ is α -stationary at $\sigma_{\mathbf{x}}$. If condition $(\mathbf{GS})$ is satisfied, then condition $(\mathbf{GS}_{\alpha})$ is satisfied for any $\alpha > 0$, and the conclusion follows. $\square$

Combining Theorem 3.1, Corollary 3.1 and Proposition 3.1, we obtain the next theorem.

Theorem 3.2 (Sequential extended extremal principle) Let $\alpha > 0$. Suppose that either X is Asplund or $\Omega_1, \dots, \Omega_n$ are convex. The collection $\{\Omega_1, \dots, \Omega_n\}$ is approximately α -stationary (approximately stationary) at $\sigma_{x_0} := \{x_0^k\} \subset \widehat{\Omega}$, $x_0^k := (x_{01}^k, \dots, x_{0n}^k)$ if and only if there exist $x^k := (x_1^k, \dots, x_n^k) \in \widehat{\Omega}$ ($k \in \mathbb{N}$) and $x_i^{*k} \in N_{\Omega_i}(x_i^k)$ ($i = 1, \dots, n, k = 1, 2, \dots$) such that $x^k - x_0^k \rightarrow 0$ as $k \rightarrow +\infty$, and condition $(\mathbf{GS}_\alpha)$ (condition $(\mathbf{GS})$) holds true.

Proof Under the assumptions, thanks to our standing convention, N is the Fréchet normal cone. The necessity follows from Theorem 3.1 (Corollary 3.1). The converse implication is a consequence of Proposition 3.1, which yields α -stationarity (stationarity) of $\{\Omega_1, \dots, \Omega_n\}$ at $\sigma_x := \{x^k\}$. It remains to notice that, thanks to Proposition 2.7, α -stationarity (stationarity) at σ_x implies approximate α -stationarity (approximate stationarity) at σ_{x_0} . $\square$

Remark 3.3 (i) Theorem 3.2 generalises and extends Theorem 1.2 as well as [10, Theorem 3.6 and Corollary 3.7] and [3, Theorem 5].

- (ii) Theorem 3.2 can be reformulated as a statement providing dual characterisations of the absence of approximate α -stationarity or approximate stationarity. These properties can be interpreted as a kind of (sequential) transversality of collections of sets; cf. [8, 9, 20, 25]. This topic is going to be explored in more detail elsewhere.
- (iii) It is easy to see from the proofs that in Lemma 3.1 and Proposition 3.1, X can be an arbitrary normed space, and the sets $\Omega_1, \dots, \Omega_n$ do not have to be closed. As a consequence, the sufficiency in Theorem 3.2 is also true in this less restrictive setting. The Banach structure of the space and closedness of the sets are only needed for the necessity part, the proof of which relies on Theorem 3.1 and Corollary 3.1.

4 Sequential necessary optimality and stationarity conditions

In this section, we consider the following minimisation problem with a single geometric constraint:

$$\text{minimize } f(x) \quad \text{subject to } x \in \Omega, \quad (P)$$

where Ω is a nonempty subset of a normed vector space X and $f : X \rightarrow \mathbb{R}_\infty$.

Let a sequence $\sigma_x := \{x^k\} \subset \Omega \cap \text{dom } f$ be given. The next two definitions extend the notion of minimising sequence. Below, $S_f(\mu)$ denotes the lower level set of f at level $\mu \in \mathbb{R}$, i.e., $S_f(\mu) := \{x \in X \mid f(x) \leq \mu\}$, and, given an $x \in X$ and an $\varepsilon > 0$, we use the notation $\zeta_{f,\Omega}(x, \varepsilon) := d(x, \Omega \cap S_f(f(x) - \varepsilon))$.

Definition 4.1 Let $\sigma_\varepsilon := \{\varepsilon^k\} \subset \mathbb{R}_+ \setminus \{0\}$ and $\varepsilon^k \downarrow 0$ as $k \rightarrow +\infty$. The sequence σ_x is

- (i) σ_ε -extremal for problem (P) if

$$\rho_{f,\Omega}(\sigma_x; \sigma_\varepsilon) := \liminf_{k \rightarrow +\infty} \zeta_{f,\Omega}(x^k, \varepsilon^k) > 0; \quad (24)$$

if $\rho_{f,\Omega}(\sigma_x; \sigma_\varepsilon) = +\infty$, we say that σ_x is globally σ_ε -extremal for problem (P);

- (ii) $(\sigma_\varepsilon, \alpha)$ -stationary for problem (P) if

$$\alpha_{f,\Omega}(\sigma_x; \sigma_\varepsilon) := \limsup_{k \rightarrow +\infty} \frac{\varepsilon^k}{\zeta_{f,\Omega}(x^k, \varepsilon^k)} < \alpha < +\infty; \quad (25)$$

if $\alpha_{f,\Omega}(\sigma_x; \sigma_\varepsilon) = 0$ (hence, (25) is satisfied with any $\alpha > 0$), we say that σ_x is σ_ε -stationary for problem (P);

(iii) approximately $(\sigma_\varepsilon, \alpha)$ -stationary for problem (P) if

$$\tilde{\alpha}_{f,\Omega}(\sigma_x; \sigma_\varepsilon) := \inf_{\sigma_u := \{u^k\} \subset \Omega, u^k - x^k \rightarrow 0, f(u^k) - f(x^k) \rightarrow 0} \alpha_{f,\Omega}(\sigma_u; \sigma_\varepsilon) < \alpha < +\infty; \quad (26)$$

if $\tilde{\alpha}_{f,\Omega}(\sigma_x; \sigma_\varepsilon) = 0$ (hence, (26) is satisfied with any $\alpha > 0$), we say that σ_x is approximately σ_ε -stationary for problem (P).

Definition 4.2 The sequence σ_x is extremal (resp., globally extremal, α -stationary, stationary, approximately α -stationary, approximately stationary) for problem (P) if it is σ_ε -extremal (resp., globally σ_ε -extremal, $(\sigma_\varepsilon, \alpha)$ -stationary, σ_ε -stationary, approximately $(\sigma_\varepsilon, \alpha)$ -stationary, approximately σ_ε -stationary) for problem (P) with some $\sigma_\varepsilon := \{\varepsilon^k\} \subset \mathbb{R}_+ \setminus \{0\}$ such that $\varepsilon^k \downarrow 0$.

Remark 4.1 The relationships between the respective properties in Definitions 4.1 and 4.2 are straightforward: σ_ε -extremality $\Rightarrow \sigma_\varepsilon$ -stationarity $\Rightarrow$ approximate σ_ε -stationarity, and extremality $\Rightarrow$ stationarity $\Rightarrow$ approximate stationarity.

Proposition 4.1 Let $\inf_\Omega f > -\infty$. If σ_x is a minimising sequence of f on Ω , i.e., $f(x^k) \rightarrow \inf_\Omega f$ as $k \rightarrow +\infty$, then it is globally extremal for problem (P).

Proof Let σ_x be a minimising sequence of f on Ω . Then there exists a sequence $\sigma_\varepsilon := \{\varepsilon^k\} \subset \mathbb{R}_+ \setminus \{0\}$ with $\varepsilon^k \downarrow 0$ such that $f(x^k) < \inf_\Omega f + \varepsilon^k$ for all $k \in \mathbb{N}$. The last inequality yields $f(x) > f(x^k) - \varepsilon^k$ for all $x \in \Omega$; hence, $\Omega \cap S_f(f(x^k) - \varepsilon^k) = \emptyset$, and consequently, $\rho_{f,\Omega}(\sigma_x; \sigma_\varepsilon) = +\infty$, i.e., σ_x is globally extremal for problem (P). $\square$

Example 4.1 Let X be a normed space, $\Omega := X$, $f(x) := -\|x\|^2$ ($x \in X$), $x^k := 0$ ($k \in \mathbb{N}$), and $\varepsilon^k \downarrow 0$. Then $S_f(-\varepsilon^k) = \{x \in X \mid \|x\| \geq \sqrt{\varepsilon^k}\}$ ($k \in \mathbb{N}$), and consequently, $\rho_{f,\Omega}(\sigma_x; \sigma_\varepsilon) = \lim_{k \rightarrow +\infty} \sqrt{\varepsilon^k} = 0$ and $\alpha_{f,\Omega}(\sigma_x; \sigma_\varepsilon) = \lim_{k \rightarrow +\infty} \frac{\varepsilon^k}{\sqrt{\varepsilon^k}} = \lim_{k \rightarrow +\infty} \sqrt{\varepsilon^k} = 0$. Hence, σ_x is not σ_ε -extremal, but is σ_ε -stationary (hence, also stationary) for problem (P).

The next proposition gives some characterisations of the properties in Definition 4.2.

Proposition 4.2 The sequence σ_x is

(i) extremal for problem (P) if and only if there exists a $\rho > 0$ such that

$$\lim_{k \rightarrow +\infty} \left(f(x^k) - \inf_{\Omega \cap B_\rho(x^k)} f \right) = 0; \quad (27)$$

(ii) globally extremal for problem (P) if and only if condition (27) is satisfied for all $\rho > 0$;

(iii) α -stationary for problem (P) if and only if

$$\alpha_{f,\Omega}^\circ(\sigma_x) := \inf_{\rho^k \downarrow 0} \limsup_{k \rightarrow +\infty} \frac{f(x^k) - \inf_{\Omega \cap B_{\rho^k}(x^k)} f}{\rho^k} < \alpha; \quad (28)$$

(iv) stationary for problem (P) if and only if $\alpha_{f,\Omega}^\circ(\sigma_x) = 0$;

(v) approximately α -stationary for problem (P) if and only if

$$\tilde{\alpha}_{f,\Omega}^\circ(\sigma_x) := \inf_{\{u^k\} \subset \Omega, u^k - x^k \rightarrow 0, f(u^k) - f(x^k) \rightarrow 0} \alpha_{f,\Omega}^\circ(\sigma_u) < \alpha;$$

(vi) approximately stationary for problem (P) if and only if $\tilde{\alpha}_{f,\Omega}^\circ(\sigma_x) = 0$.

Proof (i) Let σ_x be extremal for problem (P). By Definitions 4.1 and 4.2, there exists a sequence $\sigma_\varepsilon := \{\varepsilon^k\} \subset \mathbb{R}_+ \setminus \{0\}$ with $\varepsilon^k \downarrow 0$ such that condition (24) is satisfied, and consequently, $\zeta_{f,\Omega}(x^k, \varepsilon^k) > \rho$ for any $\rho \in (0, \rho_{f,\Omega}(\sigma_x; \sigma_\varepsilon))$ and all sufficiently large $k \in \mathbb{N}$. The latter inequality yields $\Omega \cap S_f(f(x^k) - \varepsilon^k) \cap B_\rho(x^k) = \emptyset$, and consequently, $f(x^k) < +\infty$ and $f(x) > f(x^k) - \varepsilon^k$ for all $x \in \Omega \cap B_\rho(x^k)$. Hence, $0 \leq f(x^k) - \inf_{\Omega \cap B_\rho(x^k)} f \leq \varepsilon^k$, and consequently, condition (27) is satisfied.

Conversely, let $\rho > 0$, and condition (27) be satisfied. Then there exists a sequence $\sigma_\varepsilon := \{\varepsilon^k\} \subset \mathbb{R}_+ \setminus \{0\}$ with $\varepsilon^k \downarrow 0$ such that $f(x^k) - \inf_{\Omega \cap B_\rho(x^k)} f < \varepsilon^k$ for all sufficiently large $k \in \mathbb{N}$. The latter inequality yields $f(x) > f(x^k) - \varepsilon^k$ for all $x \in \Omega \cap B_\rho(x^k)$, and consequently, $\zeta_{f,\Omega}(x^k, \varepsilon^k) \geq \rho$ ($k \in \mathbb{N}$). Hence, $\rho_{f,\Omega}(\sigma_x; \sigma_\varepsilon) \geq \rho > 0$, i.e., σ_x is σ_ε -extremal for problem (P).

- (ii) The above proof is applicable to the case of global extremality.
- (iii) Let σ_x be α -stationary for problem (P). By Definitions 4.1 and 4.2, there exists a sequence $\sigma_\varepsilon := \{\varepsilon^k\} \subset \mathbb{R}_+ \setminus \{0\}$ with $\varepsilon^k \downarrow 0$ such that condition (25) is satisfied, and consequently, there is an $\alpha' \in (0, \alpha)$ such that $\zeta_{f,\Omega}(x^k, \varepsilon^k) > \rho^k := \varepsilon^k / \alpha'$ for all sufficiently large $k \in \mathbb{N}$. As above, the latter inequality implies $0 \geq \inf_{\Omega \cap B_{\rho^k}(x^k)} f - f(x^k) \geq -\varepsilon^k = -\alpha' \rho^k$. Hence,

$$\limsup_{k \rightarrow +\infty} \frac{f(x^k) - \inf_{\Omega \cap B_{\rho^k}(x^k)} f}{\rho^k} \leq \alpha' < \alpha,$$

which yields (28).

Conversely, let condition (28) be satisfied. Then there exist a number $\alpha' \in (0, \alpha)$ and a sequence $\rho^k \downarrow 0$ such that, for all sufficiently large $k \in \mathbb{N}$, we have $f(x^k) - \inf_{\Omega \cap B_{\rho^k}(x^k)} f < \varepsilon^k := \alpha' \rho^k$, and consequently, $\zeta_{f,\Omega}(x^k, \varepsilon^k) \geq \rho^k$. Denote $\sigma_\varepsilon := \{\varepsilon^k\}$. Thus, $\varepsilon^k \downarrow 0$, and

$$\limsup_{k \rightarrow +\infty} \frac{\varepsilon^k}{\zeta_{f,\Omega}(x^k, \varepsilon^k)} \leq \alpha' < \alpha,$$

i.e., σ_x is $(\sigma_\varepsilon, \alpha)$ -stationary for problem (P).

The last three assertions are consequences of assertion (iii). $\square$

Remark 4.2 The characterisations in Proposition 4.2 give clear relationships between the extremality and stationarity properties in Definition 4.2 and the corresponding ones studied in [10, Sections 4 and 5]. Note that the definitions in [10] assume that $\{f(x^k)\}$ converges to some number μ_0 . We do not use this assumption here.

Under this superfluous assumption, a sequence $\sigma_x := \{x^k\} \subset \Omega$ is firmly inf-stationary (resp., inf-stationary, approximately inf-stationary) for problem (P) at level μ_0 in the sense of [10, Definition 4.5] if and only if some subsequence of σ_x is extremal (resp., stationary, approximately stationary) for problem (P). Thus, the definitions adopted in the current paper are more general and more specific in identifying the sequences relevant for the properties. If $x^k = \bar{x}$ for all $k \in \mathbb{N}$, the corresponding properties coincide. If, additionally, $\Omega = X$, the stationarity and approximate stationarity properties reduce to the, respectively, inf-stationarity and approximate inf-stationarity (weak inf-stationarity) studied in [22,23]. The sequential examples in [10, Examples 4.7–4.9] complement the ‘at-a-point’ Example 4.1 and provide additional illustrations of the properties.

In this paper, for brevity we do not use the ‘inf-terminology’ from [10]. This does not lead to confusion as we do not consider maximisation problems here. This way, we keep the terminology consistent with that used in Sections 2 and 3. Of course, a point of local maximum cannot be extremal in the sense of Definition 4.2 (as illustrated by Example 4.1).

To embed problem (P) into the model studied in Section 2, we consider the sets

$$\Omega_1 := \text{epi } f, \quad \Omega_2 := \Omega \times (-\infty, 0] \quad \text{and} \quad \widehat{\Omega} := \Omega_1 \times \Omega_2, \quad (29)$$

and sequences

$$x_1^k := (x^k, f(x^k)) \in \Omega_1, \quad x_2^k := (x^k, 0) \in \Omega_2 \quad \text{and} \quad \mathbf{x}^k := (x_1^k, x_2^k) \quad (k \in \mathbb{N}), \quad (30)$$

$$a_1^k := (0, 0), \quad a_2^k := (0, \varepsilon^k) \quad \text{and} \quad \mathbf{a}^k := (a_1^k, a_2^k) \quad (k \in \mathbb{N}), \quad (31)$$

where $\{\varepsilon^k\} \subset \mathbb{R}_+ \setminus \{0\}$ and $\varepsilon^k \downarrow 0$. Thus, for all $k \in \mathbb{N}$, we have $\|\mathbf{a}^k\| = \varepsilon^k$,

$$\begin{aligned} \Omega_1 - x_1^k - a_1^k &= \{(x - x^k, \mu) \mid x \in X, f(x) - f(x^k) \leq \mu\}, \\ \Omega_2 - x_2^k - a_2^k &= \{(x - x^k, \mu) \mid x \in \Omega, \mu \leq -\varepsilon^k\}, \\ (\Omega_1 - x_1^k - a_1^k) \cap (\Omega_2 - x_2^k - a_2^k) &= \{(x - x^k, \mu) \mid x \in \Omega, f(x) - f(x^k) \leq \mu \leq -\varepsilon^k\}, \\ \zeta(\mathbf{x}^k + \mathbf{a}^k) &= \max\{d(0, \Omega \cap S_f(f(x^k) - \varepsilon^k)), \varepsilon^k\} = \max\{\zeta_{f,\Omega}(x^k, \varepsilon^k), \varepsilon^k\}. \end{aligned}$$

Denote $\sigma_{\mathbf{x}} := \{\mathbf{x}^k\}$, $\sigma_{\mathbf{a}} := \{\mathbf{a}^k\}$ and $\sigma_{\varepsilon} := \{\varepsilon^k\}$. In view of the above observations, it follows from the definitions in (7), (8), (9), (24), (25) and (26) that

$$\begin{aligned} \rho(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}}) &= \liminf_{k \rightarrow +\infty} \max\{\zeta_{f,\Omega}(x^k, \varepsilon^k), \varepsilon^k\} = \liminf_{k \rightarrow +\infty} \zeta_{f,\Omega}(x^k, \varepsilon^k) = \rho_{f,\Omega}(\sigma_{\mathbf{x}}; \sigma_{\varepsilon}), \\ \alpha(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}}) &= \limsup_{k \rightarrow +\infty} \frac{\varepsilon^k}{\max\{\zeta_{f,\Omega}(x^k, \varepsilon^k), \varepsilon^k\}} \\ &= \min\left\{\limsup_{k \rightarrow +\infty} \frac{\varepsilon^k}{\zeta_{f,\Omega}(x^k, \varepsilon^k)}, 1\right\} = \min\{\alpha_{f,\Omega}(\sigma_{\mathbf{x}}; \sigma_{\varepsilon}), 1\}, \\ \widetilde{\alpha}(\sigma_{\mathbf{x}}; \sigma_{\mathbf{a}}) &= \inf_{\sigma_{\mathbf{u}} := \{\mathbf{u}^k\} \in \Omega_1 \times \Omega_2, \mathbf{u}^k - \mathbf{x}^k \rightarrow 0} \alpha(\sigma_{\mathbf{u}}; \sigma_{\mathbf{a}}) \\ &\leq \inf_{\substack{\sigma_{\mathbf{u}} := \{\mathbf{u}^k\} \in \Omega, \mathbf{u}^k - \mathbf{x}^k \rightarrow 0, \\ f(\mathbf{u}^k) - f(\mathbf{x}^k) \rightarrow 0}} \min\{\alpha_{f,\Omega}(\sigma_{\mathbf{u}}; \sigma_{\varepsilon}), 1\} = \min\{\widetilde{\alpha}_{f,\Omega}(\sigma_{\mathbf{x}}; \sigma_{\varepsilon}), 1\}. \end{aligned}$$

We can now formulate relationships between the properties in Definitions 4.1 and 4.2, and the corresponding ones in Definitions 2.1 and 2.2.

Proposition 4.3 *Let $\Omega_1, \Omega_2, x_1^k, x_2^k, a_1^k, a_2^k, \mathbf{x}^k$ and $\mathbf{a}^k$ ($k \in \mathbb{N}$) be given by (29), (30) and (31), and $\alpha \in (0, 1)$. The sequence $\sigma_{\mathbf{x}}$ is σ_{ε} -extremal (resp., $(\sigma_{\varepsilon}, \alpha)$ -stationary, σ_{ε} -stationary) for problem (P) if and only if $\{\Omega_1, \Omega_2\}$ is extremal (resp., α -stationary, stationary) at $\sigma_{\mathbf{x}}$ with respect to $\sigma_{\mathbf{a}}$.*

If $\sigma_{\mathbf{x}}$ is approximately $(\sigma_{\varepsilon}, \alpha)$ -stationary (resp., approximately σ_{ε} -stationary) for problem (P), then $\{\Omega_1, \Omega_2\}$ is approximately α -stationary (resp., approximately stationary) at $\sigma_{\mathbf{x}}$ with respect to $\sigma_{\mathbf{a}}$.

As a consequence, if $\sigma_{\mathbf{x}}$ is extremal (resp., α -stationary, stationary, approximately α -stationary, approximately stationary) for problem (P), then $\{\Omega_1, \Omega_2\}$ is extremal (resp., α -stationary, stationary, approximately α -stationary, approximately stationary) at $\sigma_{\mathbf{x}}$.

In the next theorem and the rest of the paper, we suppose that X is Banach, f is lower semicontinuous and Ω is closed. As a consequence, the sets Ω_1 and Ω_2 given by (29) are closed. Recall our standing convention that the subdifferential and normal cone generic notations ∂ and N stand for ∂^C and N^C if X is a general Banach space, and for ∂^F and N^F if X is Asplund.

Theorem 4.1 Suppose that σ_x is approximately stationary (particularly, stationary or extremal) for problem (P). Then there exist $(x_1^k, \mu_1^k) \in \text{epi } f$, $(x_1^{*k}, v_1^k) \in N_{\text{epi } f}(x_1^k, \mu_1^k)$, $x_2^k \in \Omega$, $\mu_2^k \leq 0$, $x_2^{*k} \in N_\Omega(x_2^k)$, and $v_2^k \geq 0$ such that $v_2^k \mu_2^k = 0$ ($k \in \mathbb{N}$),

$$x_1^k - x^k \rightarrow 0, x_2^k - x^k \rightarrow 0, \mu_1^k - f(x^k) \rightarrow 0, \mu_2^k \rightarrow 0 \text{ as } k \rightarrow +\infty, \quad (32)$$

$$x_1^{*k} + x_2^{*k} \rightarrow 0, \quad v_1^k + v_2^k \rightarrow 0 \text{ as } k \rightarrow +\infty, \quad (33)$$

$$\|x_1^{*k}\| + |v_1^k| = 1 \quad (k \in \mathbb{N}). \quad (34)$$

Proof By Proposition 4.3, the pair $\{\Omega_1, \Omega_2\}$ given by (29) is approximately stationary at σ_x . The statement follows from Corollary 3.1 applied with $n = 2$: there exist $(x_1^k, \mu_1^k) \in \text{epi } f$, $(x_2^k, \mu_2^k) \in \Omega_2$, $(x_1^{*k}, v_1^k) \in N_{\text{epi } f}(x_1^k, \mu_1^k)$, and $(x_2^{*k}, v_2^k) \in N_{\Omega_2}(x_2^k, \mu_2^k)$ such that conditions (32) and (33) are satisfied, and $\|((x_1^{*k}, v_1^k), (x_2^{*k}, v_2^k))\| = 1$ ($k \in \mathbb{N}$). Then $x_2^{*k} \in N_\Omega(x_2^k)$, $v_2^k \geq 0$, $v_2^k \mu_2^k = 0$ ($k \in \mathbb{N}$), and

$$\lim_{k \rightarrow +\infty} (\|x_1^{*k}\| + |v_1^k|) = \lim_{k \rightarrow +\infty} (\|x_2^{*k}\| + v_2^k) = \frac{1}{2}.$$

We can assume without loss of generality that $(x_1^{*k}, v_1^k) \neq 0$ for all $k \in \mathbb{N}$. Replacing normal vectors (x_1^{*k}, v_1^k) and (x_2^{*k}, v_2^k) by the ‘stretched’ normal vectors $(x_1^{*k}, v_1^k)/(\|x_1^{*k}\| + |v_1^k|)$ and $(x_2^{*k}, v_2^k)/(\|x_2^{*k}\| + v_2^k)$, respectively, while keeping the original notation, we can ensure condition (34). Conditions (33) remain true. $\square$

Remark 4.3 Since vector (x_1^{*k}, v_1^k) in Theorem 4.1 is normal to $\text{epi } f$, it obviously holds $v_1^k \leq 0$ and $v_1^k(\mu_1^k - f(x_1^k)) = 0$ for all $k \in \mathbb{N}$.

Theorem 4.1 yields the following theorem-of-alternative type statement.

Corollary 4.1 Suppose that σ_x is approximately stationary (particularly, stationary or extremal) for problem (P). Then one of the following assertions holds true:

- (i) there exist an $M > 0$, $x_1^k \in X$, $x_2^k \in \Omega$, $x_1^{*k} \in \partial f(x_1^k)$ and $x_2^{*k} \in N_\Omega(x_2^k) \cap (M\mathbb{B}_{X^*})$ ($k \in \mathbb{N}$) such that $f(x_1^k) - f(x^k) \rightarrow 0$,

$$x_1^k - x^k \rightarrow 0, x_2^k - x^k \rightarrow 0 \quad \text{and} \quad x_1^{*k} + x_2^{*k} \rightarrow 0 \text{ as } k \rightarrow +\infty; \quad (35)$$

- (ii) there exist $(x_1^k, \mu_1^k) \in \text{epi } f$, $x_2^k \in \Omega$, $\mu_2^k \leq 0$, $(x_1^{*k}, v_1^k) \in N_{\text{epi } f}(x_1^k, \mu_1^k)$, $x_2^{*k} \in N_\Omega(x_2^k)$, and $v_2^k \geq 0$ ($k \in \mathbb{N}$) such that $v_2^k \mu_2^k = 0$ ($k \in \mathbb{N}$), conditions (35) are satisfied,

$$\mu_1^k - f(x^k) \rightarrow 0, \mu_2^k \rightarrow 0, v_1^k \rightarrow 0, v_2^k \rightarrow 0 \quad \text{and} \quad \|x_1^{*k}\| = \|x_2^{*k}\| = 1 \quad (k \in \mathbb{N}). \quad (36)$$

Proof By Theorem 4.1, there exist $(x_1^k, \mu_1^k) \in \text{epi } f$, $x_2^k \in \Omega$, $\mu_2^k \leq 0$, $(x_1^{*k}, v_1^k) \in N_{\text{epi } f}(x_1^k, \mu_1^k)$, $x_2^{*k} \in N_\Omega(x_2^k)$, and $v_2^k \geq 0$ ($k \in \mathbb{N}$) such that $v_2^k \mu_2^k = 0$ ($k \in \mathbb{N}$), and conditions (32)–(34) are satisfied. In the rest of the proof, without further mentioning, all claims hold ‘for all sufficiently large $k \in \mathbb{N}$ ’. Denote $\gamma := \liminf_{k \rightarrow +\infty} |v_1^k|$. Two cases are possible leading to the two assertions in the theorem.

$\gamma > 0$. It follows from (33) that $\liminf_{k \rightarrow +\infty} v_2^k = \gamma > 0$. Hence, $v_1^k < 0$ and $v_2^k > 0$, and consequently, $\mu_1^k = f(x_1^k)$ and $\mu_2^k = 0$. Thus, $f(x_1^k) - f(x^k) \rightarrow 0$, and conditions (35) are satisfied. Denote $M := \gamma^{-1}$. By (33) and (34),

$$\limsup_{k \rightarrow +\infty} \|x_1^{*k}\|/|v_1^k| < M \quad \text{and} \quad \limsup_{k \rightarrow +\infty} \|x_2^{*k}\|/v_2^k < M.$$

It remains to observe that $x_1^{*k} := x_1^{*k}/|v_1^k| \in \partial f(x_1^k)$ and $x_2^{*k} := x_2^{*k}/v_2^k \in N_\Omega(x_2^k) \cap (M\mathbb{B}_{X^*})$.

$\gamma = 0$. Passing to subsequences while keeping the original notation, thanks to (33) and (34), we have $v_1^k \uparrow 0$, $v_2^k \downarrow 0$, $\|x_1^{*k}\| \rightarrow 1$ and $\|x_2^{*k}\| \rightarrow 1$. Hence, $|v_1^k|/\|x_1^{*k}\| \rightarrow 0$ and $v_2^k/\|x_2^{*k}\| \rightarrow 0$. Scaling the normal vectors, we can assume without loss of generality that $\|x_1^{*k}\| = \|x_2^{*k}\| = 1$. Thus, conditions (36) are satisfied. $\square$

- Remark 4.4* (i) Thanks to Proposition 4.1, the conclusions of Theorem 4.1 and Corollary 4.1 hold true, in particular, if $\inf_{\Omega} f > -\infty$, and σ_x is a conventional minimising sequence. In this case, Theorem 4.1 and Corollary 4.1 improve [10, Theorem 5.2 and Corollary 5.3].
- (ii) The arguments in the proof of Corollary 4.1(i) lead to a seemingly stronger than $x_1^{*k} \in \partial f(x_1^k)$ condition $x_1^{*k} \in \partial f(x_1^k) \cap (M\mathbb{B}_{X^*})$. It is not difficult to see that, thanks to $\|x_2^{*k}\| < M$ and $x_1^{*k} + x_2^{*k} \rightarrow 0$, the latter condition is implied by $x_1^{*k} \in \partial f(x_1^k)$ (with possibly a larger M).
- (iii) Corollary 4.1(i) yields a kind of multiplier rule:

$$d(0, \partial f(x_1^k) + N_{\Omega}(x_2^k) \cap (M\mathbb{B}_{X^*})) \rightarrow 0 \text{ as } k \rightarrow +\infty.$$

The limiting version (at an abstract infinity point) of this rule in finite dimensions is given in [14, Theorem 6.1].

- (iv) Part (ii) of Corollary 4.1 corresponds to ‘singular’ behaviour of f on Ω with the normal vectors (x_1^{*k}, v_1^k) to the epigraph of f being ‘almost horizontal’. If $\mu_1^k > f(x_1^k)$, then $v_1^k = 0$ and x_1^{*k} is normal to $\text{dom } f$ at x_1^k . *Qualification conditions* can be used to exclude the singular behaviour in Corollary 4.1(ii), thus, leaving the multiplier rule type assertion in part (i) as the only possibility. The simplest sufficient conditions of this kind are: 1) f is Lipschitz continuous, and 2) $\Omega = X$. In the first case, condition $v_1^k \rightarrow 0$ implies $x_1^{*k} \rightarrow 0$, while in the second, we have $x_2^{*k} = 0$. Thus, in both cases one of the equalities in (36) is violated. The mentioned sufficient conditions are rather strong and far from being necessary. More subtle qualification conditions can be found in [10, Section 5].

Example 4.2 Let X be a Banach space, $\Omega := X$, $f(x) := -\|x\|^2$ ($x \in X$), and $x^k := 0$. The ‘sequence’ $\sigma_x := \{x^k\}$ is stationary for problem (P) (see Example 4.1). In view of Remark 4.4(iv), assertion (i) in Corollary 4.1 must hold true. Indeed, take $M := 1$, $x_1^k = x_2^k = u^k := 0$. Then $f(x_1^k) = 0$, $x_1^{*k} = \nabla f(x_1^k) = 0$, $N_{\Omega}(x_2^k) = \{0\}$, and consequently, $x_2^{*k} = 0$. Hence, assertion (i) in Corollary 4.1 holds true.

Example 4.3 Let $X := \mathbb{R}$, $\Omega := \mathbb{R}_+$, $f(x) := 0$ if $x \leq 0$ and $f(x) := -x$ if $x > 0$. Note that f is Lipschitz continuous. Let $x^k := 0$ ($k \in \mathbb{N}$). Choose any $\{\varepsilon^k\} \subset \mathbb{R}_+ \setminus \{0\}$ such that $\varepsilon^k \downarrow 0$ and denote $\sigma_{\varepsilon} := \{\varepsilon^k\}$. Then $S_f(f(x^k) - \varepsilon^k) = \{x \in \mathbb{R} \mid x \geq \varepsilon^k\}$, $\zeta_{f,\Omega}(x^k, \varepsilon^k) = \varepsilon^k$, $\alpha_{f,\Omega}(\sigma_x; \sigma_{\varepsilon}) = \lim_{k \rightarrow +\infty} \frac{\varepsilon^k}{\varepsilon^k} = 1$. Since $\{\varepsilon^k\}$ is arbitrary, σ_x is not stationary for problem (P).

Let $u^k := -\sqrt{\varepsilon^k}$ ($k \in \mathbb{N}$) and $\sigma_u := \{u^k\}$. Then $\zeta_{f,\Omega}(u^k, \varepsilon^k) = \varepsilon^k + \sqrt{\varepsilon^k}$, and consequently, $\alpha_{f,\Omega}(\sigma_u; \sigma_{\varepsilon}) = \lim_{k \rightarrow +\infty} \frac{\varepsilon^k}{\varepsilon^k + \sqrt{\varepsilon^k}} = 0$. Hence, σ_u is σ_{ε} -stationary, and σ_x is approximately σ_{ε} -stationary for problem (P). In view of Remark 4.4(iv), assertion (i) in Corollary 4.1 must hold true. Indeed, take $M := 1$, $x_1^k := -(\varepsilon^k)^2$ and $x_2^k := 0$. Then $f(x_1^k) = 0$, $x_1^{*k} = \nabla f(x_1^k) = 0$ and $x_2^{*k} := 0 \in N_{\Omega}(x_2^k)$. Conditions (35) are trivially satisfied. Hence, assertion (i) in Corollary 4.1 holds true.

Some sequential examples can be found in [10, Section 5].

5 Conclusions

We establish some refinements of the sequential concepts of extremality and stationarity of a collection of sets and the *sequential extremal principle* introduced recently in [10]. The model adopted in the current paper extends and improves the one studied in [10] along several lines as discussed in the Introduction. In particular, the definitions adopted in the current paper are more specific in identifying the sequences relevant for the extremality and stationarity properties. We discuss the properties corresponding to fixed sequences of translations. This line of research is new even in the conventional ‘at-a-point’ setting. This leads to improved

sequential dual necessary conditions with some additional restrictions on the dual variables and also sheds some new light on extremality/stationarity settings in general, especially on the role of the vectors/sequences of translations.

We show that the sequential (as well as conventional) extremality and approximate stationarity properties possess certain stability, while the (non-approximate) stationarity does not.

In the dual statements, including the sequential extended extremal principle, we employ certain generalised separation conditions (**GS**) and (**GS** _{α}) as well as a complementary primal-dual condition (**PD**) providing additional restrictions on the associated dual vectors. The conditions are formulated in terms of Clarke (in general Banach spaces) or Fréchet (in Asplund spaces) normals in an approximate/fuzzy form. Imposing certain sequential normal compactness assumptions (which are automatically satisfied in finite dimensions), one can formulate limiting versions of the conditions in terms of certain types of limiting normal cones. The sequential extended extremal principle can be reformulated as a statement providing dual characterisations of the absence of approximate stationarity. This property can be interpreted as a kind of (sequential) transversality of collections of sets. The last two topics are going to be explored in more detail elsewhere.

To illustrate the model, we consider a constrained minimisation problem in which the minimal value is not necessarily attained and deduce stronger than in [10] sequential optimality and stationarity conditions for more general types of stationary sequences.

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Declarations

Conflict of interest. The authors have no competing interests to declare that are relevant to the content of this article.

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